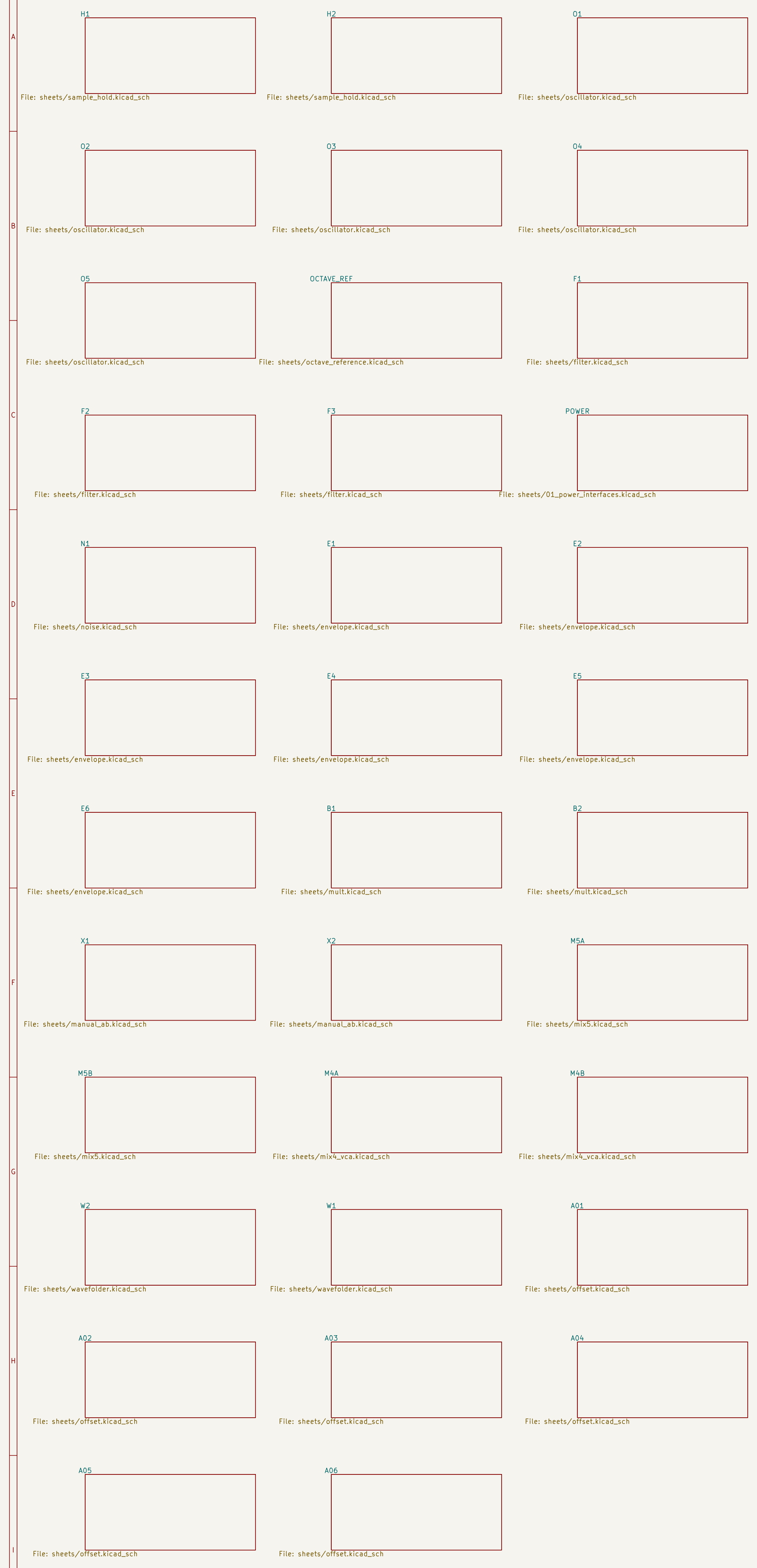

H1
File: sheets/sample_hold.kicad_sch
H2
File: sheets/sample_hold.kicad_sch
O1
File: sheets/oscillator.kicad_sch
O2
File: sheets/oscillator.kicad_sch
O3
File: sheets/oscillator.kicad_sch
O4
File: sheets/oscillator.kicad_sch
O5
File: sheets/oscillator.kicad_sch
OCTAVE_REF
File: sheets/octave_reference.kicad_sch
F1
File: sheets/filter.kicad_sch
F2
File: sheets/filter.kicad_sch
F3
File: sheets/filter.kicad_sch
POWER
File: sheets/01_power_interfaces.kicad_sch
N1
File: sheets/noise.kicad_sch
E1
File: sheets/envelope.kicad_sch
E2
File: sheets/envelope.kicad_sch
E3
File: sheets/envelope.kicad_sch
E4
File: sheets/envelope.kicad_sch
E5
File: sheets/envelope.kicad_sch
E6
File: sheets/envelope.kicad_sch
B1
File: sheets/mult.kicad_sch
B2
File: sheets/mult.kicad_sch
X1
File: sheets/manual_ab.kicad_sch
X2
File: sheets/manual_ab.kicad_sch
M5A
File: sheets/mix5.kicad_sch
M5B
File: sheets/mix5.kicad_sch
M4A
File: sheets/mix4_vca.kicad_sch
M4B
File: sheets/mix4_vca.kicad_sch
W2
File: sheets/wavefolder.kicad_sch
W1
File: sheets/wavefolder.kicad_sch
A01
File: sheets/offset.kicad_sch
A02
File: sheets/offset.kicad_sch
A03
File: sheets/offset.kicad_sch
A04
File: sheets/offset.kicad_sch
A05
File: sheets/offset.kicad_sch
A06
File: sheets/offset.kicad_sch

Sheet: /
File: zudo–osc–hole–field.kicad_sch

Title:

Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/36

U101
U102A
U102B
U102C
U102D
U102E
OPA4197IPWR
U103A
U103B
U103C
U103D
U103E
OPA4197IDR
U104A
U104B
U104C
U104D
U104E
TMUX7412FRRPR
U106A
U106B
U106C
U106D
U106E
OPA4196IDR
U107A
U107B
U107C
LM393BIDR
U108A
U108B
U108C
U108D
U108E
U108F
U108G
SN74HC14DR
U110A
U110B
U110C
U110D
U110E
U111A
U111B
U111C
U111D
U111E
U111F
U111G
U112A
U112B
U112C
U112D
U112E
U113A
U113B
U113C
U113D
U113E
U116A
U116B
U116C
U116D
U116E
U117A
U117B
U117C
U117D
U117E
U118A
U118B
U118C
U118D
U118E
U119A
U119B
U119C
U119D
U119E
RV801
DH101
DH102
DH117
D509
D609
D709
BAS16W-QX
APHHS1005QWF/D
BAT54S,215
AGND
+12V
-12V
+5V
TRIGGER_TIP
RAW_HELD
SLEW_STORAGE
SLEW_BUFFERED
REF_5V
REF_N5V
GATE_REF
TRIGGER_PROTECTED
TRIGGER_BUFFERED
IN_BUFFERED
IN_PROTECTED
TRIGGER_GATE
TRIGGER_OR
BUTTON_ACTIVE
SAMPLE_PULSE
TIMING_RCX
TIMING_C
HOLD_CAP
SLEW_POT_IN
OUT_TIP
CONTACT
C5195c0_PRE_OUT
Cdc0c61_RAW_5V
Cdc0c61_INV
Cdc0c61_GATE_RAW
Cdded02_ENABLE
C382003_ENABLE
C8a1aae_DEBOUNCE
C267eb3_SENSE
C307276_GATE_SENSE
C307276_THRESH
C307276_CMP_OUT
Ce0e057_SENSE
C07a7c2_SENSE
C07a7c2_DRIVE
C07a7c2_LED_A
C07a7c2_LED_K
Ca073c3_SENSE
Ca073c3_DRIVE
Ca073c3_LED_A
Ca073c3_LED_K
C372c4e_SENSE
C372c4e_DRIVE
C372c4e_LED_A
C372c4e_LED_K
C1979a1_DRIVE
C1979a1_FB
C1979a1_ISO_MID
reference_generator_J_H1_TRIGGER___A1_IO_UNUSED1
reference_generator_J_H1_TRIGGER___A2_IO_UNUSED
reference_generator_J_H1_TRIGGER___A3_IO_UNUSED4
high_impedance_input_J_H1_IN___A_IO_UNUSED4
magnitude_indicator_J_H1_TRIGGER_mag___A_IO_UNUSED4
magnitude_indicator_J_H1_IN_mag___A_IO_UNUSED4
slew_island_C_H1_SLEW___PRE_IO_UNUSED4
slew_island_C_H1_SLEW___POST_IO_UNUSED4
precision_output_J_H1_OUT___A_IO_UNUSED4
magnitude_indicator_J_H1_OUT_mag___A_IO_UNUSED4
DNP
reservation

Sheet: /H1/
File: sample_hold.kicad_sch

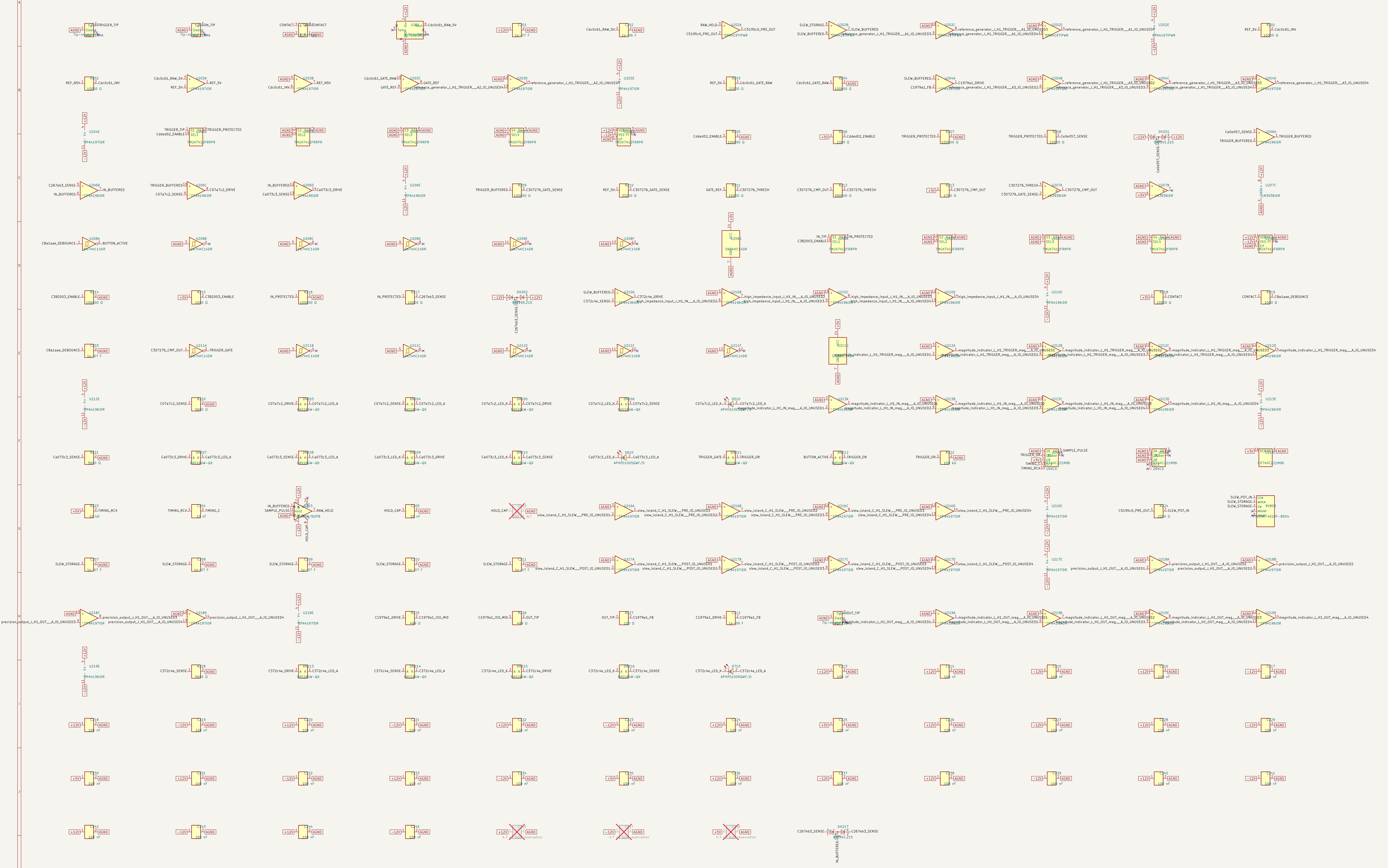
Title:

Size: A1 Date:

Rev:

KiCad E.D.A. 10.0.6

Id: 2/36



Sheet: /H2/
File: sample_hold.kicad_sch

Title:

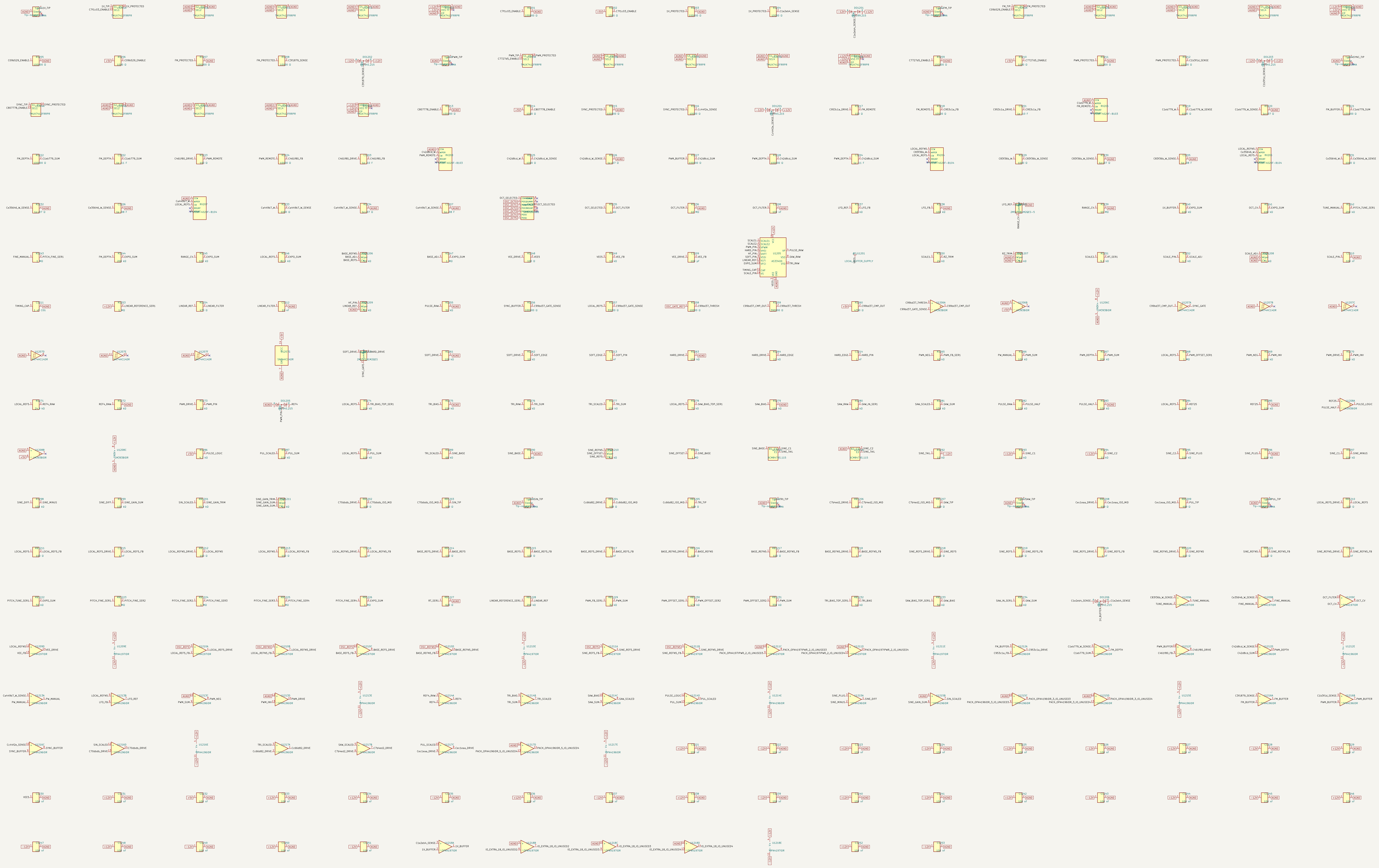
Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/36

Sheet: /01/
File: oscillator.kicad_sch

Title:

Size: A0 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 4/36



Sheet: /02/
File: oscillator.kicad_sch

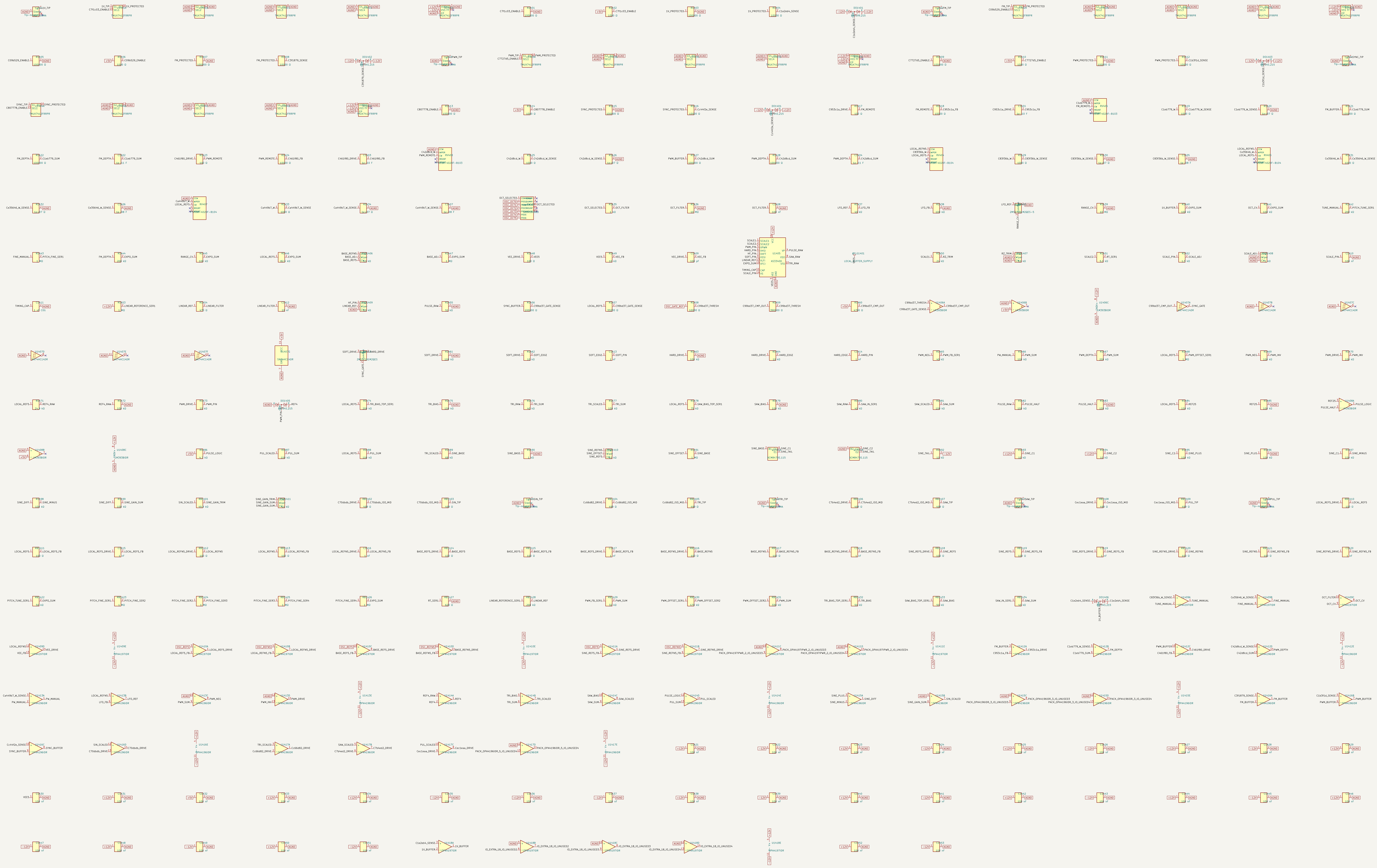
Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 5/36

Sheet: /03/
File: oscillator.kicad_sch

Title:

Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 6/36

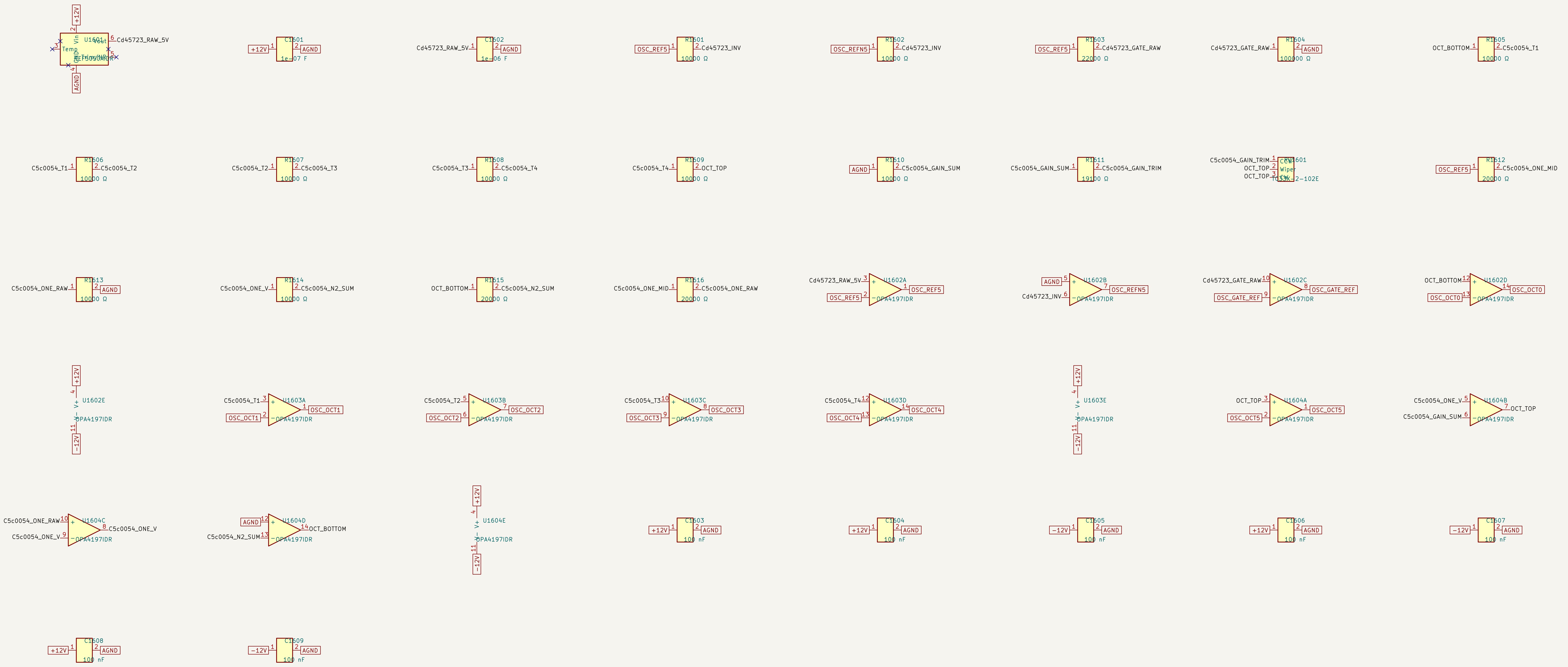


Sheet: /04/
File: oscillator.kicad_sch

Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 7/36

Sheet: /05/
File: oscillator.kicad_sch
Title:
Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 8/36


U1601
REF5050AIDR
Vin
Vout
Temp
Trim/NR
GND
+12V
AGND
Cd45723_RAW_5V
C1601
+12V
AGND
1e-07 F
C1602
Cd45723_RAW_5V
AGND
1e-06 F
R1601
OSC_REF5
Cd45723_INV
10000 Ω
R1602
OSC_REFN5
Cd45723_INV
10000 Ω
R1603
OSC_REF5
Cd45723_GATE_RAW
22000 Ω
R1604
Cd45723_GATE_RAW
AGND
100000 Ω
R1605
OCT_BOTTOM
C5c0054_T1
10000 Ω
R1606
C5c0054_T1
C5c0054_T2
10000 Ω
R1607
C5c0054_T2
C5c0054_T3
10000 Ω
R1608
C5c0054_T3
C5c0054_T4
10000 Ω
R1609
C5c0054_T4
OCT_TOP
10000 Ω
R1610
AGND
C5c0054_GAIN_SUM
10000 Ω
R1611
C5c0054_GAIN_SUM
C5c0054_GAIN_TRIM
19100 Ω
RV1601
C5c0054_GAIN_TRIM
OCT_TOP
OCT_TOP
CCW
Wiper
CW
TC33X-2-102E
R1612
OSC_REF5
C5c0054_ONE_MID
20000 Ω
R1613
C5c0054_ONE_RAW
AGND
10000 Ω
R1614
C5c0054_ONE_V
C5c0054_N2_SUM
10000 Ω
R1615
OCT_BOTTOM
C5c0054_N2_SUM
20000 Ω
R1616
C5c0054_ONE_MID
C5c0054_ONE_RAW
20000 Ω
U1602A
Cd45723_RAW_5V
OSC_REF5
OSC_REF5
OPA4197IDR
U1602B
AGND
Cd45723_INV
OSC_REFN5
OPA4197IDR
U1602C
Cd45723_GATE_RAW
OSC_GATE_REF
OSC_GATE_REF
OPA4197IDR
U1602D
OCT_BOTTOM
OSC_OCT0
OSC_OCT0
OPA4197IDR
U1602E
+12V
-12V
V+
V-
OPA4197IDR
U1603A
C5c0054_T1
OSC_OCT1
OSC_OCT1
OPA4197IDR
U1603B
C5c0054_T2
OSC_OCT2
OSC_OCT2
OPA4197IDR
U1603C
C5c0054_T3
OSC_OCT3
OSC_OCT3
OPA4197IDR
U1603D
C5c0054_T4
OSC_OCT4
OSC_OCT4
OPA4197IDR
U1603E
+12V
-12V
V+
V-
OPA4197IDR
U1604A
OCT_TOP
OSC_OCT5
OSC_OCT5
OPA4197IDR
U1604B
C5c0054_ONE_V
C5c0054_GAIN_SUM
OCT_TOP
OPA4197IDR
U1604C
C5c0054_ONE_RAW
C5c0054_ONE_V
C5c0054_ONE_V
OPA4197IDR
U1604D
AGND
C5c0054_N2_SUM
OCT_BOTTOM
OPA4197IDR
U1604E
+12V
-12V
V+
V-
OPA4197IDR
C1603
+12V
AGND
100 nF
C1604
+12V
AGND
100 nF
C1605
-12V
AGND
100 nF
C1606
+12V
AGND
100 nF
C1607
-12V
AGND
100 nF
C1608
+12V
AGND
100 nF
C1609
-12V
AGND
100 nF

Sheet: /OCTAVE_REF/
File: octave_reference.kicad_sch

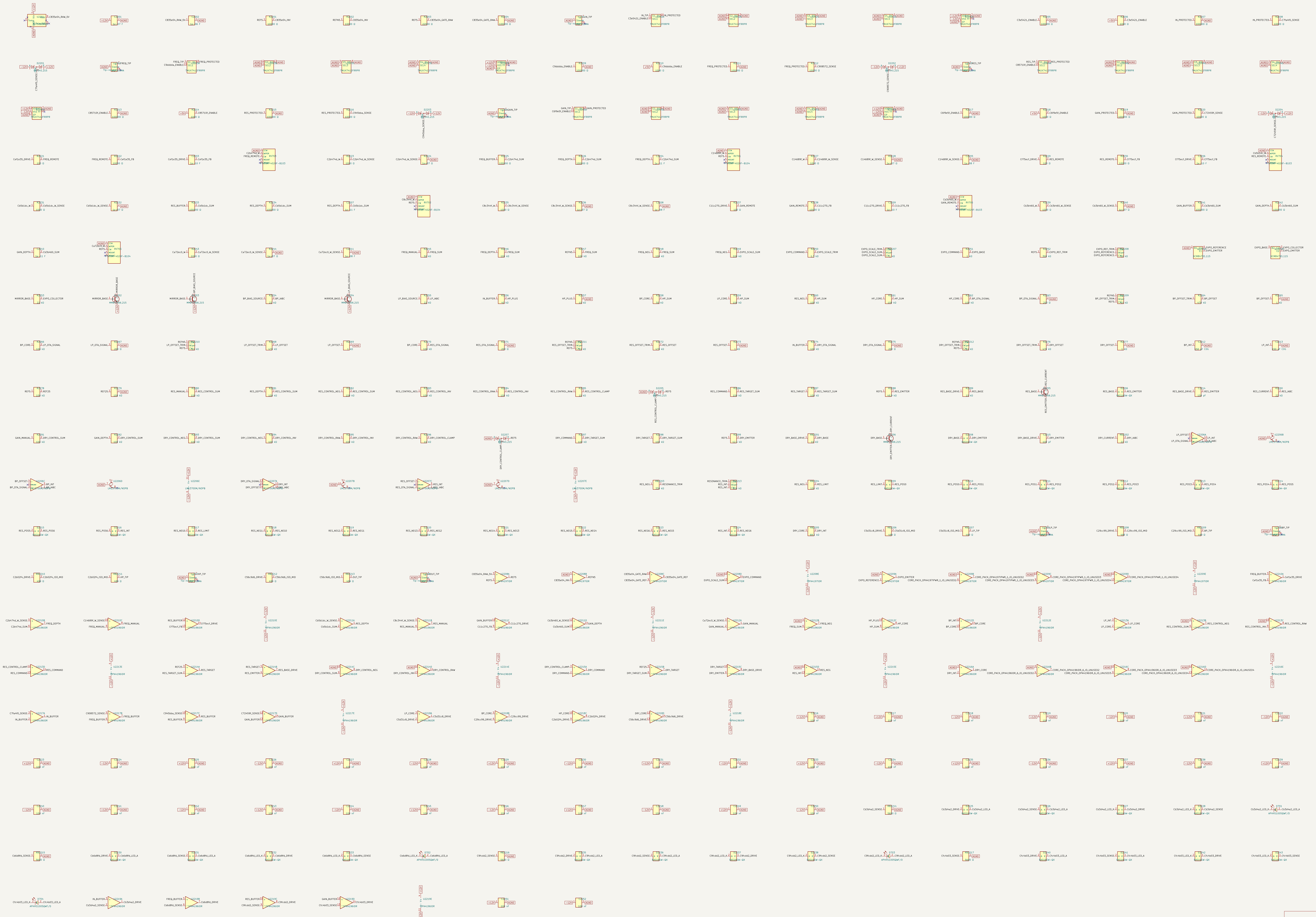
Title:

Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 9/36

Sheet: /F1/
File: filter.kicad_sch

Title:

Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 10/36

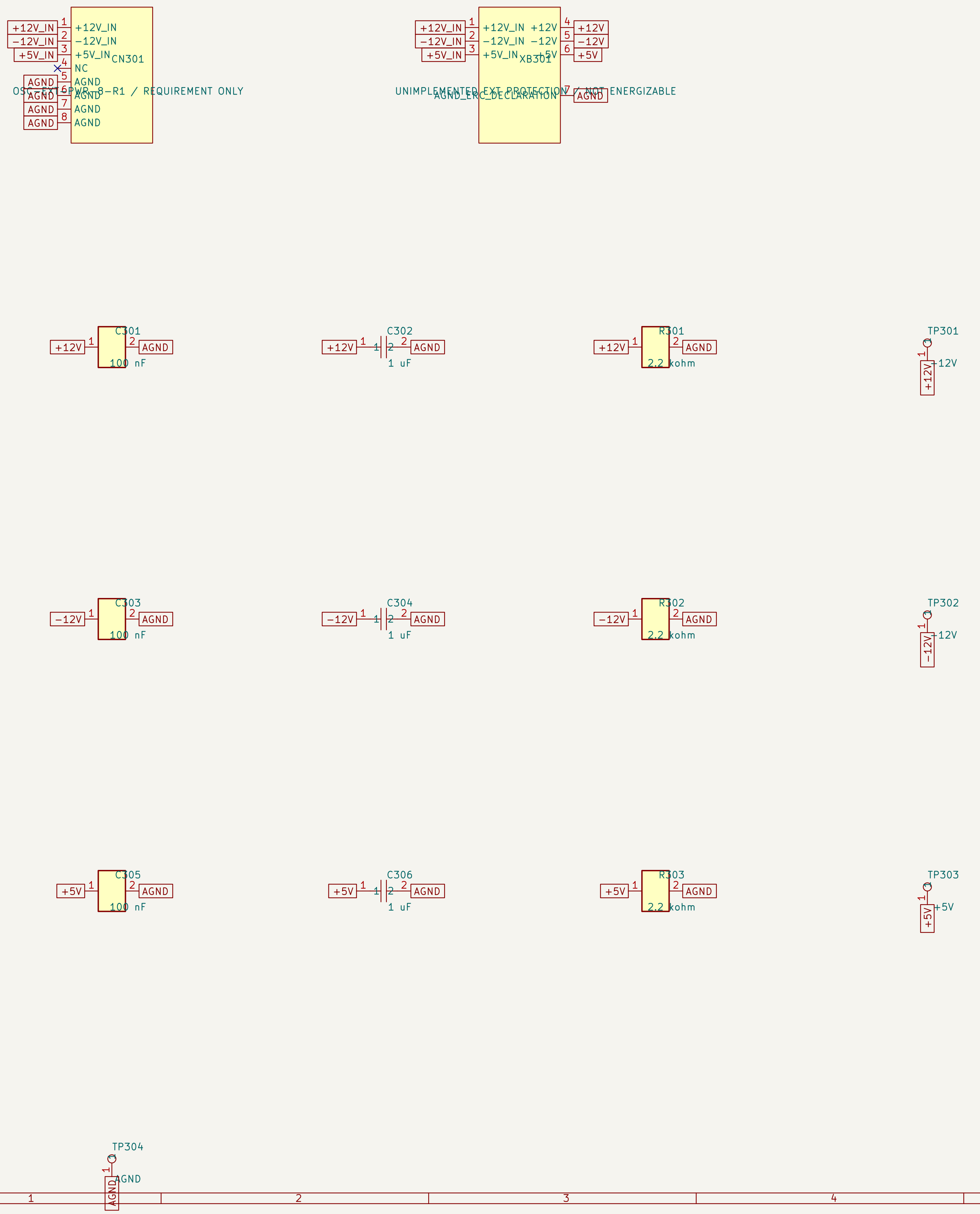


Sheet: /F2/
File: filter.kicad_sch

Title:

Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 11/36

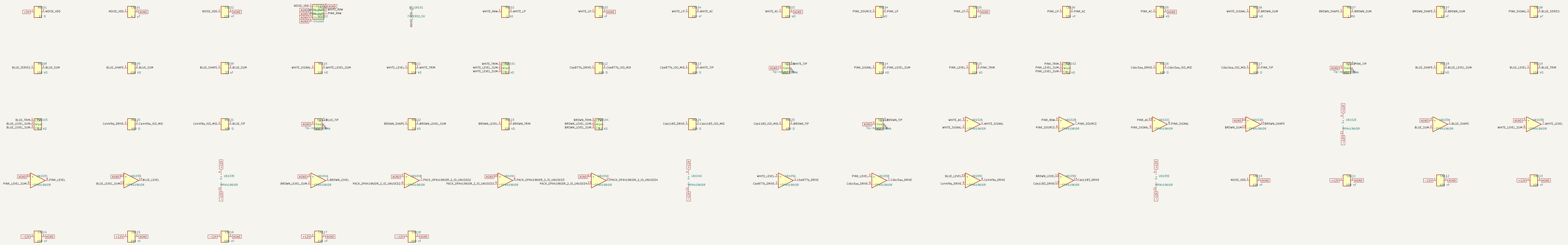
Sheet: /F3/
File: filter.kicad_sch
Title:
Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 12/36


+12V_IN
−12V_IN
+5V_IN
NC
AGND
AGND
AGND
AGND
+12V_IN
−12V_IN
+5V_IN
AGND
AGND
AGND
AGND
CN301
OSC-EXT-PWR-8-R1 / REQUIREMENT ONLY
+12V_IN
−12V_IN
+5V_IN
+12V_IN
−12V_IN
+5V_IN
+12V
−12V
+5V
+12V
−12V
+5V
XB301
UNIMPLEMENTED EXT PROTECTION / NOT ENERGIZABLE
AGND_ERC_DECLARATION
AGND
C301
+12V
AGND
100 nF
C302
+12V
AGND
1 uF
R301
+12V
AGND
2.2 kohm
TP301
+12V
+12V
C303
−12V
AGND
100 nF
C304
−12V
AGND
1 uF
R302
−12V
AGND
2.2 kohm
TP302
−12V
−12V
C305
+5V
AGND
100 nF
C306
+5V
AGND
1 uF
R303
+5V
AGND
2.2 kohm
TP303
+5V
+5V
TP304
AGND
AGND

Sheet: /POWER/
File: 01_power_interfaces.kicad_sch

Title:

Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 13/36


+5V R6101 NOISE_VDD 22 Ω
NOISE_VDD C6101 AGND 4.7 µF
NOISE_VDD C6102 AGND 100 nF
NOISE_VDD U6101 +5V/0V AGND WHITE_RAW PINK_RAW NOISE2 Unused
#FLG6101 NOISE_VDD FILTERED_5V
WHITE_RAW R6102 WHITE_LP 1 kΩ
WHITE_LP C6103 AGND 100 nF
WHITE_LP C6104 WHITE_AC 100 nF
WHITE_AC R6103 AGND 100 kΩ
PINK_SOURCE R6104 PINK_LP 1 kΩ
PINK_LP C6105 AGND 10 nF
PINK_LP C6106 PINK_AC 100 nF
PINK_AC R6105 AGND 100 kΩ
WHITE_SIGNAL R6106 BROWN_SUM 100 kΩ
BROWN_SHAPE R6107 BROWN_SUM 1 MΩ
BROWN_SHAPE C6107 BROWN_SUM 10 nF
PINK_SIGNAL C6108 BLUE_SERIES 100 pF
BLUE_SERIES R6108 BLUE_SUM 100 kΩ
BLUE_SHAPE R6109 BLUE_SUM 100 kΩ
BLUE_SHAPE C6109 BLUE_SUM 33 pF
WHITE_SIGNAL R6110 WHITE_LEVEL_SUM 100 kΩ
WHITE_LEVEL R6111 WHITE_TRIM 100 kΩ
WHITE_TRIM WHITE_LEVEL_SUM WHITE_LEVEL_SUM RV6101 CCW Wiper CW 10 kΩ
Cbe877b_DRIVE R6112 Cbe877b_ISO_MID 499 Ω
Cbe877b_ISO_MID R6113 WHITE_TIP 499 Ω
AGND Tip Sleeve WHITE_TIP TN Tip-normal 10MA
PINK_SIGNAL R6114 PINK_LEVEL_SUM 100 kΩ
PINK_LEVEL R6115 PINK_TRIM 100 kΩ
PINK_TRIM PINK_LEVEL_SUM PINK_LEVEL_SUM RV6102 CCW Wiper CW 10 kΩ
Cdbc5aa_DRIVE R6116 Cdbc5aa_ISO_MID 499 Ω
Cdbc5aa_ISO_MID R6117 PINK_TIP 499 Ω
AGND Tip Sleeve PINK_TIP TN Tip-normal 10MA
BLUE_SHAPE R6118 BLUE_LEVEL_SUM 10 kΩ
BLUE_LEVEL R6119 BLUE_TRIM 100 kΩ
BLUE_TRIM BLUE_LEVEL_SUM BLUE_LEVEL_SUM RV6103 CCW Wiper CW 10 kΩ
Ce44f9a_DRIVE R6120 Ce44f9a_ISO_MID 499 Ω
Ce44f9a_ISO_MID R6121 BLUE_TIP 499 Ω
AGND Tip Sleeve BLUE_TIP TN Tip-normal 10MA
BROWN_SHAPE R6122 BROWN_LEVEL_SUM 20 kΩ
BROWN_LEVEL R6123 BROWN_TRIM 100 kΩ
BROWN_TRIM BROWN_LEVEL_SUM BROWN_LEVEL_SUM RV6104 CCW Wiper CW 10 kΩ
Cde1183_DRIVE R6124 Cde1183_ISO_MID 499 Ω
Cde1183_ISO_MID R6125 BROWN_TIP 499 Ω
AGND Tip Sleeve BROWN_TIP TN Tip-normal 10MA
WHITE_AC WHITE_SIGNAL U6102A OPA4196IDR WHITE_SIGNAL
PINK_RAW PINK_SOURCE U6102B OPA4196IDR PINK_SOURCE
PINK_AC PINK_SIGNAL U6102C OPA4196IDR PINK_SIGNAL
AGND BROWN_SUM U6102D OPA4196IDR BROWN_SHAPE
+12V U6102E OPA4196IDR −12V
AGND BLUE_SUM U6103A OPA4196IDR BLUE_SHAPE
AGND WHITE_LEVEL_SUM U6103B OPA4196IDR WHITE_LEVEL
AGND PINK_LEVEL_SUM U6103C OPA4196IDR PINK_LEVEL
AGND BLUE_LEVEL_SUM U6103D OPA4196IDR BLUE_LEVEL
+12V U6103E OPA4196IDR −12V
AGND BROWN_LEVEL_SUM U6104A OPA4196IDR BROWN_LEVEL
AGND PACK_OPA4196IDR_2_IO_UNUSED2 U6104B OPA4196IDR PACK_OPA4196IDR_2_IO_UNUSED2
AGND PACK_OPA4196IDR_2_IO_UNUSED3 U6104C OPA4196IDR PACK_OPA4196IDR_2_IO_UNUSED3
AGND PACK_OPA4196IDR_2_IO_UNUSED4 U6104D OPA4196IDR PACK_OPA4196IDR_2_IO_UNUSED4
+12V U6104E OPA4196IDR −12V
WHITE_LEVEL Cbe877b_DRIVE U6105A OPA4196IDR Cbe877b_DRIVE
PINK_LEVEL Cdbc5aa_DRIVE U6105B OPA4196IDR Cdbc5aa_DRIVE
BLUE_LEVEL Ce44f9a_DRIVE U6105C OPA4196IDR Ce44f9a_DRIVE
BROWN_LEVEL Cde1183_DRIVE U6105D OPA4196IDR Cde1183_DRIVE
+12V U6105E OPA4196IDR −12V
NOISE_VDD C6110 AGND 100 nF
+12V C6111 AGND 100 nF
−12V C6112 AGND 100 nF
+12V C6113 AGND 100 nF
−12V C6114 AGND 100 nF
+12V C6115 AGND 100 nF
−12V C6116 AGND 100 nF
+12V C6117 AGND 100 nF
−12V C6118 AGND 100 nF

Sheet: /N1/
File: noise.kicad_sch

Title:

Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 14/36

Sheet: /E1/
File: envelope.kicad_sch

Title:

Size: A0 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 15/36

Sheet: /E2/
File: envelope.kicad_sch

Title:

Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 16/36

Sheet: /E3/
File: envelope.kicad_sch

Title:

Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 17/36

Sheet: /E4/
File: envelope.kicad_sch

Title:

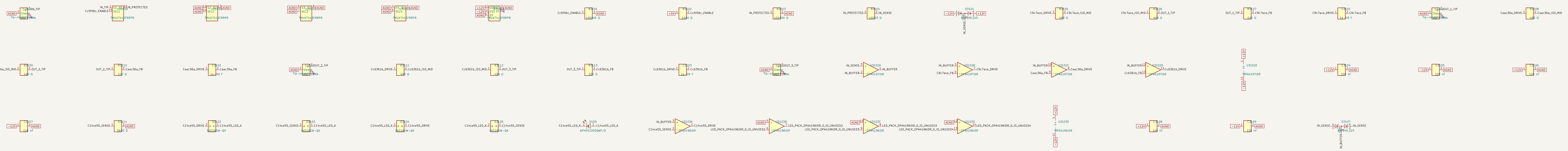
Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 18/36

Sheet: /E5/
File: envelope.kicad_sch
Title:
Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 19/36

Sheet: /E6/
File: envelope.kicad_sch

Title:

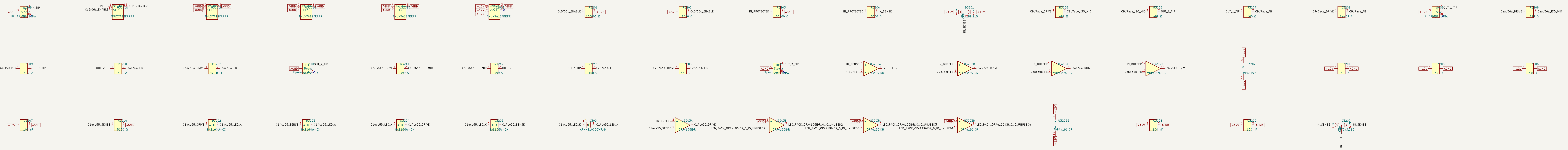
Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 20/36


AGND
S
Tip
Sleeve
T
IN_TIP
TN
Tip-normal
IN_TIP
Cc5f06c_ENABLE
S1
D1
SEL1
U3101A
IN_PROTECTED
TMUX7412FRRPR
AGND
S2
D2
SEL2
U3101B
TMUX7412FRRPR
S3
D3
SEL3
U3101C
TMUX7412FRRPR
S4
D4
SEL4
U3101D
TMUX7412FRRPR
+12V
-12V
VDD
VSS
EP
GND
FF
U3101E
TMUX7412FRRPR
R3101
Cc5f06c_ENABLE
100000 Ω
R3102
+5V
Cc5f06c_ENABLE
1000 Ω
R3103
IN_PROTECTED
100000 Ω
R3104
IN_PROTECTED
IN_SENSE
10000 Ω
D3101
A
K
COM
BAV199,215
IN_SENSE
R3105
C9c7ace_DRIVE
C9c7ace_ISO_MID
499 Ω
R3106
C9c7ace_ISO_MID
OUT_1_TIP
499 Ω
R3107
OUT_1_TIP
C9c7ace_FB
100 Ω
C3101
C9c7ace_DRIVE
C9c7ace_FB
1e-09 F
OUT_1_TIP
R3108
Caac36a_DRIVE
Caac36a_ISO_MID
499 Ω
R3109
Caac36a_ISO_MID
OUT_2_TIP
499 Ω
R3110
OUT_2_TIP
Caac36a_FB
100 Ω
C3102
Caac36a_DRIVE
Caac36a_FB
1e-09 F
OUT_2_TIP
R3111
Cc6361b_DRIVE
Cc6361b_ISO_MID
499 Ω
R3112
Cc6361b_ISO_MID
OUT_3_TIP
499 Ω
R3113
OUT_3_TIP
Cc6361b_FB
100 Ω
C3103
Cc6361b_DRIVE
Cc6361b_FB
1e-09 F
OUT_3_TIP
U3102A
IN_SENSE
IN_BUFFER
IN_BUFFER
OPA4197IDR
U3102B
IN_BUFFER
C9c7ace_FB
C9c7ace_DRIVE
OPA4197IDR
U3102C
IN_BUFFER
Caac36a_FB
Caac36a_DRIVE
OPA4197IDR
U3102D
IN_BUFFER
Cc6361b_FB
Cc6361b_DRIVE
OPA4197IDR
U3102E
V+
V-
OPA4197IDR
C3104
100 nF
C3105
100 nF
C3106
100 nF
C3107
100 nF
R3114
C14ce55_SENSE
5600 Ω
D3102
C14ce55_DRIVE
C14ce55_LED_A
BAS16GW-QX
D3103
C14ce55_SENSE
C14ce55_LED_A
BAS16GW-QX
D3104
C14ce55_LED_K
C14ce55_DRIVE
BAS16GW-QX
D3105
C14ce55_LED_K
C14ce55_SENSE
BAS16GW-QX
D109
C14ce55_LED_K
C14ce55_LED_A
APHHS1005QWF/D
U3103A
IN_BUFFER
C14ce55_SENSE
C14ce55_DRIVE
OPA4196IDR
U3103B
LED_PACK_OPA4196IDR_0_IO_UNUSED2
OPA4196IDR
U3103C
LED_PACK_OPA4196IDR_0_IO_UNUSED3
OPA4196IDR
U3103D
LED_PACK_OPA4196IDR_0_IO_UNUSED4
OPA4196IDR
U3103E
OPA4196IDR
C3108
100 nF
C3109
100 nF
D3107
IN_SENSE
IN_SENSE
IN_BUFFER
BAT54S,215

Sheet: /B1/
File: mult.kicad_sch

Title:

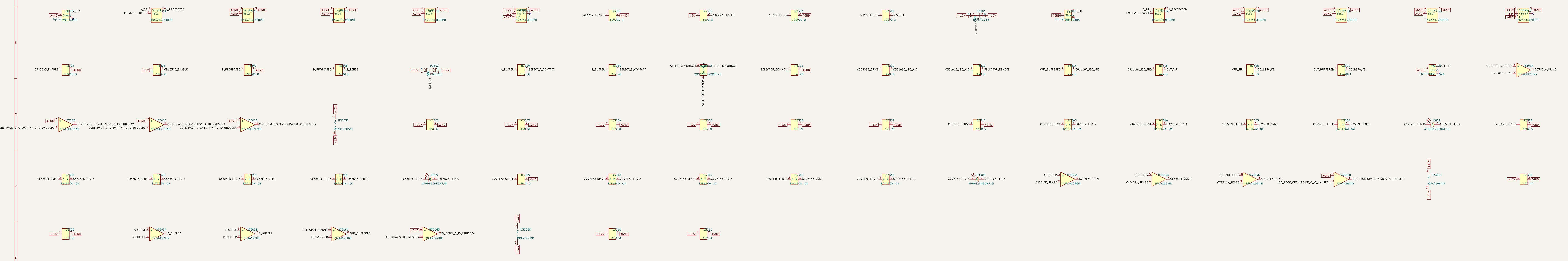
Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 21/36


AGND
Tip
Sleeve
Tip-normal
IN_TIP
IN_TIP
Cc5f06c_ENABLE
U3201A
S1
D1
SEL1
IN_PROTECTED
TMUX7412FRRPR
AGND
AGND
U3201B
S2
D2
SEL2
AGND
TMUX7412FRRPR
AGND
AGND
U3201C
S3
D3
SEL3
AGND
TMUX7412FRRPR
AGND
AGND
U3201D
S4
D4
SEL4
AGND
TMUX7412FRRPR
+12V
-12V
AGND
VDD
VSS
EP
GND
FF
AGND
TMUX7412FRRPR
R3201
Cc5f06c_ENABLE
AGND
100000 Ω
R3202
+5V
Cc5f06c_ENABLE
1000 Ω
R3203
IN_PROTECTED
AGND
100000 Ω
R3204
IN_PROTECTED
IN_SENSE
10000 Ω
D3201
-12V
+12V
IN_SENSE
BAV199,215
R3205
C9c7ace_DRIVE
C9c7ace_ISO_MID
499 Ω
R3206
C9c7ace_ISO_MID
OUT_1_TIP
499 Ω
R3207
OUT_1_TIP
C9c7ace_FB
100 Ω
C3201
C9c7ace_DRIVE
C9c7ace_FB
1e-09 F
AGND
Tip
Sleeve
Tip-normal
OUT_1_TIP
R3208
Caac36a_DRIVE
Caac36a_ISO_MID
499 Ω
R3209
Caac36a_ISO_MID
OUT_2_TIP
499 Ω
R3210
OUT_2_TIP
Caac36a_FB
100 Ω
C3202
Caac36a_DRIVE
Caac36a_FB
1e-09 F
AGND
Tip
Sleeve
Tip-normal
OUT_2_TIP
R3211
Cc6361b_DRIVE
Cc6361b_ISO_MID
499 Ω
R3212
Cc6361b_ISO_MID
OUT_3_TIP
499 Ω
R3213
OUT_3_TIP
Cc6361b_FB
100 Ω
C3203
Cc6361b_DRIVE
Cc6361b_FB
1e-09 F
AGND
Tip
Sleeve
Tip-normal
OUT_3_TIP
U3202A
IN_SENSE
IN_BUFFER
IN_BUFFER
OPA4197IDR
U3202B
IN_BUFFER
C9c7ace_FB
C9c7ace_DRIVE
OPA4197IDR
U3202C
IN_BUFFER
Caac36a_FB
Caac36a_DRIVE
OPA4197IDR
U3202D
IN_BUFFER
Cc6361b_FB
Cc6361b_DRIVE
OPA4197IDR
U3202E
+12V
-12V
OPA4197IDR
C3204
+12V
AGND
100 nF
C3205
-12V
AGND
100 nF
C3206
+12V
AGND
100 nF
C3207
-12V
AGND
100 nF
R3214
C14ce55_SENSE
AGND
5600 Ω
D3202
C14ce55_DRIVE
C14ce55_LED_A
BAS16GW-QX
D3203
C14ce55_SENSE
C14ce55_LED_A
BAS16GW-QX
D3204
C14ce55_LED_K
C14ce55_DRIVE
BAS16GW-QX
D3205
C14ce55_LED_K
C14ce55_SENSE
BAS16GW-QX
D309
C14ce55_LED_K
C14ce55_LED_A
APHHS1005QWF/D
U3203A
IN_BUFFER
C14ce55_SENSE
C14ce55_DRIVE
OPA4196IDR
U3203B
AGND
LED_PACK_OPA4196IDR_0_IO_UNUSED2
LED_PACK_OPA4196IDR_0_IO_UNUSED2
OPA4196IDR
U3203C
AGND
LED_PACK_OPA4196IDR_0_IO_UNUSED3
LED_PACK_OPA4196IDR_0_IO_UNUSED3
OPA4196IDR
U3203D
AGND
LED_PACK_OPA4196IDR_0_IO_UNUSED4
LED_PACK_OPA4196IDR_0_IO_UNUSED4
OPA4196IDR
U3203E
+12V
-12V
OPA4196IDR
C3208
+12V
AGND
100 nF
C3209
-12V
AGND
100 nF
D3207
IN_SENSE
IN_SENSE
IN_BUFFER
BAT54S,215

Sheet: /B2/
File: mult.kicad_sch

Title:

Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 22/36


AGND
Tip
Sleeve
Tip-normal
WQP518MA
A_TIP
A_TIP
Cadd797_ENABLE
U3301A
SEL1
A_PROTECTED
TMUX7412FRRPR
U3301B
SEL2
AGND
TMUX7412FRRPR
U3301C
SEL3
AGND
TMUX7412FRRPR
U3301D
SEL4
AGND
TMUX7412FRRPR
+12V
-12V
AGND
VDD
VSS
EP
GND
FF
AGND
TMUX7412FRRPR
Cadd797_ENABLE
R3301
AGND
100000 Ω
+5V
R3302
Cadd797_ENABLE
1000 Ω
A_PROTECTED
R3303
AGND
100000 Ω
A_PROTECTED
R3304
A_SENSE
10000 Ω
-12V
D3301
+12V
BAT54S,215
A_SENSE
AGND
Tip
Sleeve
Tip-normal
WQP518MA
B_TIP
B_TIP
C9a8343_ENABLE
U3302A
SEL1
B_PROTECTED
TMUX7412FRRPR
U3302B
SEL2
AGND
TMUX7412FRRPR
U3302C
SEL3
AGND
TMUX7412FRRPR
U3302D
SEL4
AGND
TMUX7412FRRPR
+12V
-12V
AGND
VDD
VSS
EP
GND
FF
AGND
TMUX7412FRRPR
C9a8343_ENABLE
R3305
AGND
100000 Ω
+5V
R3306
C9a8343_ENABLE
1000 Ω
B_PROTECTED
R3307
AGND
100000 Ω
B_PROTECTED
R3308
B_SENSE
10000 Ω
-12V
D3302
+12V
BAT54S,215
B_SENSE
A_BUFFER
R3309
SELECT_A_CONTACT
2.2 kΩ
B_BUFFER
R3310
SELECT_B_CONTACT
2.2 kΩ
SELECT_A_CONTACT
SELECT_B_CONTACT
2MS1T1B1M2QES-5
SELECTOR_COMMON
SELECTOR_COMMON
R3311
AGND
10 MΩ
C33d018_DRIVE
R3312
C33d018_ISO_MID
499 Ω
C33d018_ISO_MID
R3313
SELECTOR_REMOTE
499 Ω
OUT_BUFFERED
R3314
C61b194_ISO_MID
499 Ω
C61b194_ISO_MID
R3315
OUT_TIP
499 Ω
OUT_TIP
R3316
C61b194_FB
100 Ω
OUT_BUFFERED
C3301
C61b194_FB
1e-09 F
AGND
Tip
Sleeve
Tip-normal
WQP518MA
OUT_TIP
SELECTOR_COMMON
U3303A
C33d018_DRIVE
C33d018_DRIVE
OPA4197IPWR
AGND
CORE_PACK_OPA4197IPWR_0_IO_UNUSED2
U3303B
CORE_PACK_OPA4197IPWR_0_IO_UNUSED2
OPA4197IPWR
AGND
CORE_PACK_OPA4197IPWR_0_IO_UNUSED3
U3303C
CORE_PACK_OPA4197IPWR_0_IO_UNUSED3
OPA4197IPWR
AGND
CORE_PACK_OPA4197IPWR_0_IO_UNUSED4
U3303D
CORE_PACK_OPA4197IPWR_0_IO_UNUSED4
OPA4197IPWR
+12V
U3303E
OPA4197IPWR
-12V
+12V
C3302
AGND
100 nF
-12V
C3303
AGND
100 nF
+12V
C3304
AGND
100 nF
-12V
C3305
AGND
100 nF
+12V
C3306
AGND
100 nF
-12V
C3307
AGND
100 nF
C025c3f_SENSE
R3317
AGND
5600 Ω
C025c3f_DRIVE
D3303
C025c3f_LED_A
BAS16GW-QX
C025c3f_SENSE
D3304
C025c3f_LED_A
BAS16GW-QX
C025c3f_LED_K
D3305
C025c3f_DRIVE
BAS16GW-QX
C025c3f_LED_K
D3306
C025c3f_SENSE
BAS16GW-QX
C025c3f_LED_K
D809
C025c3f_LED_A
APHHS1005QWF/D
Cc6c62b_SENSE
R3318
AGND
5600 Ω
Cc6c62b_DRIVE
D3308
Cc6c62b_LED_A
BAS16GW-QX
Cc6c62b_SENSE
D3309
Cc6c62b_LED_A
BAS16GW-QX
Cc6c62b_LED_K
D3310
Cc6c62b_DRIVE
BAS16GW-QX
Cc6c62b_LED_K
D3311
Cc6c62b_SENSE
BAS16GW-QX
Cc6c62b_LED_K
D909
Cc6c62b_LED_A
APHHS1005QWF/D
C7971de_SENSE
R3319
AGND
5600 Ω
C7971de_DRIVE
D3313
C7971de_LED_A
BAS16GW-QX
C7971de_SENSE
D3314
C7971de_LED_A
BAS16GW-QX
C7971de_LED_K
D3315
C7971de_DRIVE
BAS16GW-QX
C7971de_LED_K
D3316
C7971de_SENSE
BAS16GW-QX
C7971de_LED_K
D1009
C7971de_LED_A
APHHS1005QWF/D
A_BUFFER
C025c3f_SENSE
U3304A
C025c3f_DRIVE
OPA4196IDR
B_BUFFER
Cc6c62b_SENSE
U3304B
Cc6c62b_DRIVE
OPA4196IDR
OUT_BUFFERED
C7971de_SENSE
U3304C
C7971de_DRIVE
OPA4196IDR
AGND
LED_PACK_OPA4196IDR_0_IO_UNUSED4
U3304D
LED_PACK_OPA4196IDR_0_IO_UNUSED4
OPA4196IDR
+12V
U3304E
OPA4196IDR
-12V
+12V
C3308
AGND
100 nF
-12V
C3309
AGND
100 nF
A_SENSE
A_BUFFER
U3305A
A_BUFFER
OPA4197IDR
B_SENSE
B_BUFFER
U3305B
B_BUFFER
OPA4197IDR
SELECTOR_REMOTE
C61b194_FB
U3305C
OUT_BUFFERED
OPA4197IDR
AGND
IO_EXTRA_5_IO_UNUSED4
U3305D
IO_EXTRA_5_IO_UNUSED4
OPA4197IDR
+12V
U3305E
OPA4197IDR
-12V
+12V
C3310
AGND
100 nF
-12V
C3311
AGND
100 nF

Sheet: /X1/
File: manual_ab.kicad_sch

Title:

Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 23/36

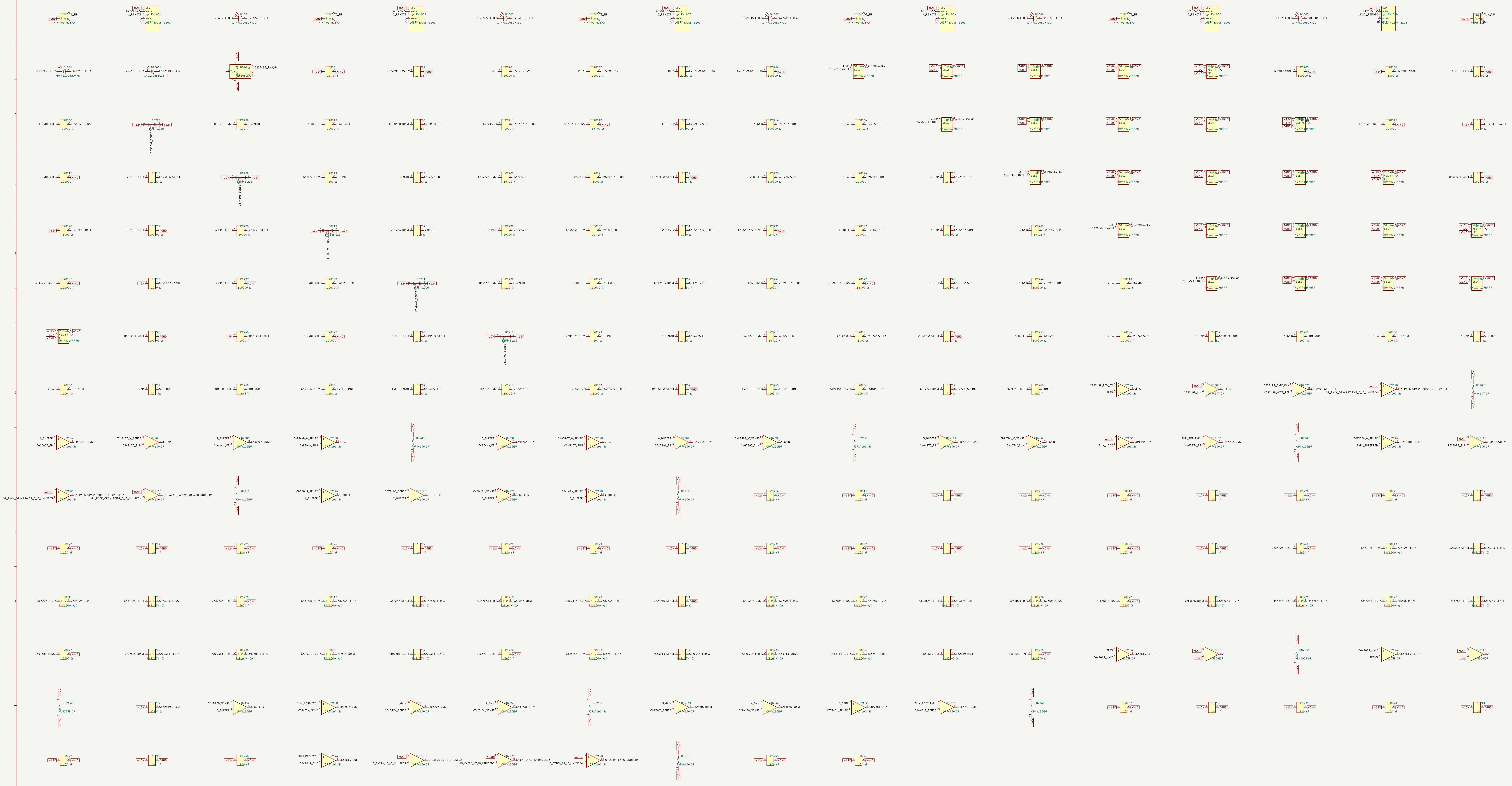
A
B
C
D
E
A_TIP
AGND
S
Tip
Sleeve
Tip-normal
TN
A_TIP
Cadd797_ENABLE
A_PROTECTED
S1
D1
SEL1
U3401A
TMUX7412FRRPR
AGND
S2
D2
SEL2
U3401B
TMUX7412FRRPR
S3
D3
SEL3
U3401C
TMUX7412FRRPR
S4
D4
SEL4
U3401D
TMUX7412FRRPR
+12V
-12V
VDD
VSS
EP
GND
FF
U3401E
TMUX7412FRRPR
R3401
Cadd797_ENABLE
100000 Ω
R3402
+5V
1000 Ω
R3403
A_PROTECTED
100000 Ω
R3404
A_SENSE
10000 Ω
D3401
BAT54S,215
COM
B_TIP
B_PROTECTED
C9a8343_ENABLE
U3402A
U3402B
U3402C
U3402D
U3402E
R3405
C9a8343_ENABLE
100000 Ω
R3406
1000 Ω
R3407
B_PROTECTED
100000 Ω
R3408
B_SENSE
10000 Ω
D3402
BAT54S,215
R3409
A_BUFFER
SELECT_A_CONTACT
2.2 kΩ
R3410
B_BUFFER
SELECT_B_CONTACT
2.2 kΩ
SELECT_A_CONTACT
SELECT_B_CONTACT
SELECTOR_COMMON
Common
2MS1T1B1M2QES-5
R3411
10 MΩ
R3412
C33d018_DRIVE
C33d018_ISO_MID
499 Ω
R3413
SELECTOR_REMOTE
499 Ω
R3414
OUT_BUFFERED
C61b194_ISO_MID
499 Ω
R3415
OUT_TIP
499 Ω
R3416
C61b194_FB
100 Ω
C3401
1e-09 F
OUT_TIP
U3403A
OPA4197IPWR
U3403B
CORE_PACK_OPA4197IPWR_0_IO_UNUSED2
U3403C
CORE_PACK_OPA4197IPWR_0_IO_UNUSED3
U3403D
CORE_PACK_OPA4197IPWR_0_IO_UNUSED4
U3403E
V+
V-
C3402
100 nF
C3403
C3404
C3405
C3406
C3407
R3417
C025c3f_SENSE
5600 Ω
D3403
C025c3f_DRIVE
C025c3f_LED_A
BAS16GW-QX
D3404
D3405
C025c3f_LED_K
D3406
D810
APHHS1005QWF/D
R3418
Cc6c62b_SENSE
D3408
Cc6c62b_DRIVE
Cc6c62b_LED_A
D3409
D3410
Cc6c62b_LED_K
D3411
D910
R3419
C7971de_SENSE
D3413
C7971de_DRIVE
C7971de_LED_A
D3414
D3415
C7971de_LED_K
D3416
D1010
U3404A
OPA4196IDR
U3404B
U3404C
U3404D
LED_PACK_OPA4196IDR_0_IO_UNUSED4
U3404E
C3408
C3409
U3405A
A_SENSE
OPA4197IDR
U3405B
U3405C
C61b194_FB
U3405D
IO_EXTRA_5_IO_UNUSED4
U3405E
C3410
C3411
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24

Sheet: /X2/
File: manual_ab.kicad_sch
Title:
Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 24/36

Sheet: /M5A/
File: mix5.kicad_sch

Title:

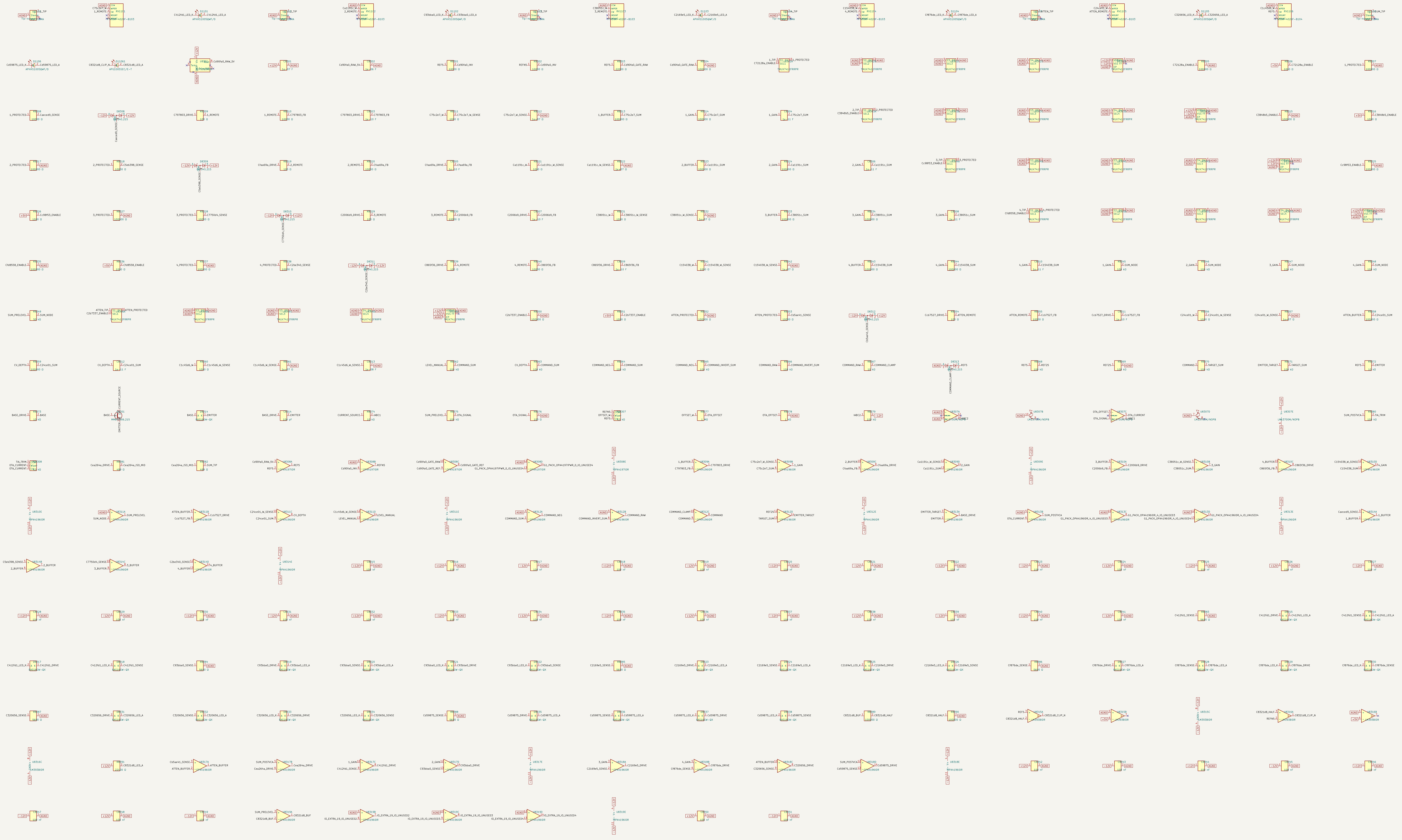
Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 25/36



Sheet: /M5B/
File: mix5.kicad_sch

Title:

Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 26/36



Sheet: /M4A/
File: mix4_vca.kicad_sch

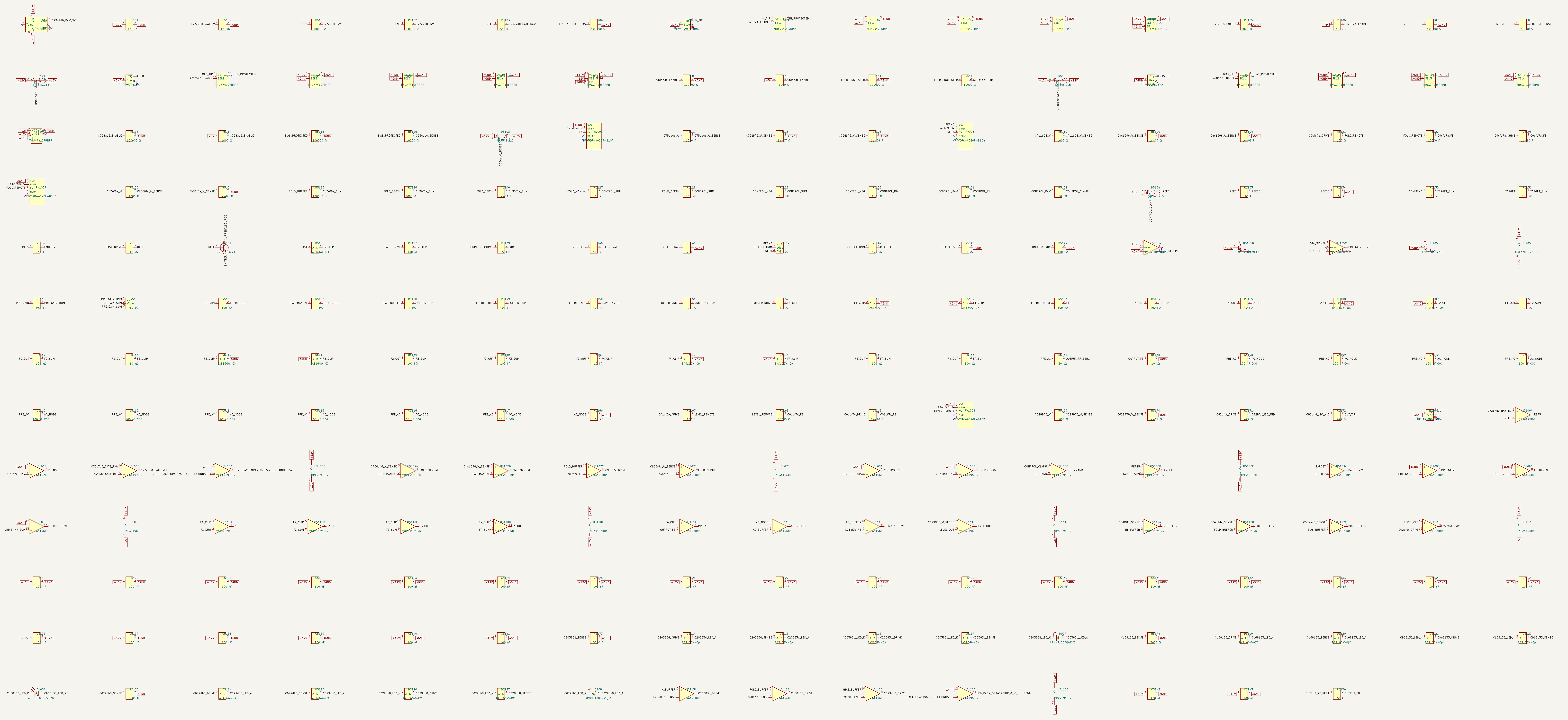
Title:

Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 27/36

Sheet: /M4B/
File: mix4_vca.kicad_sch

Title:

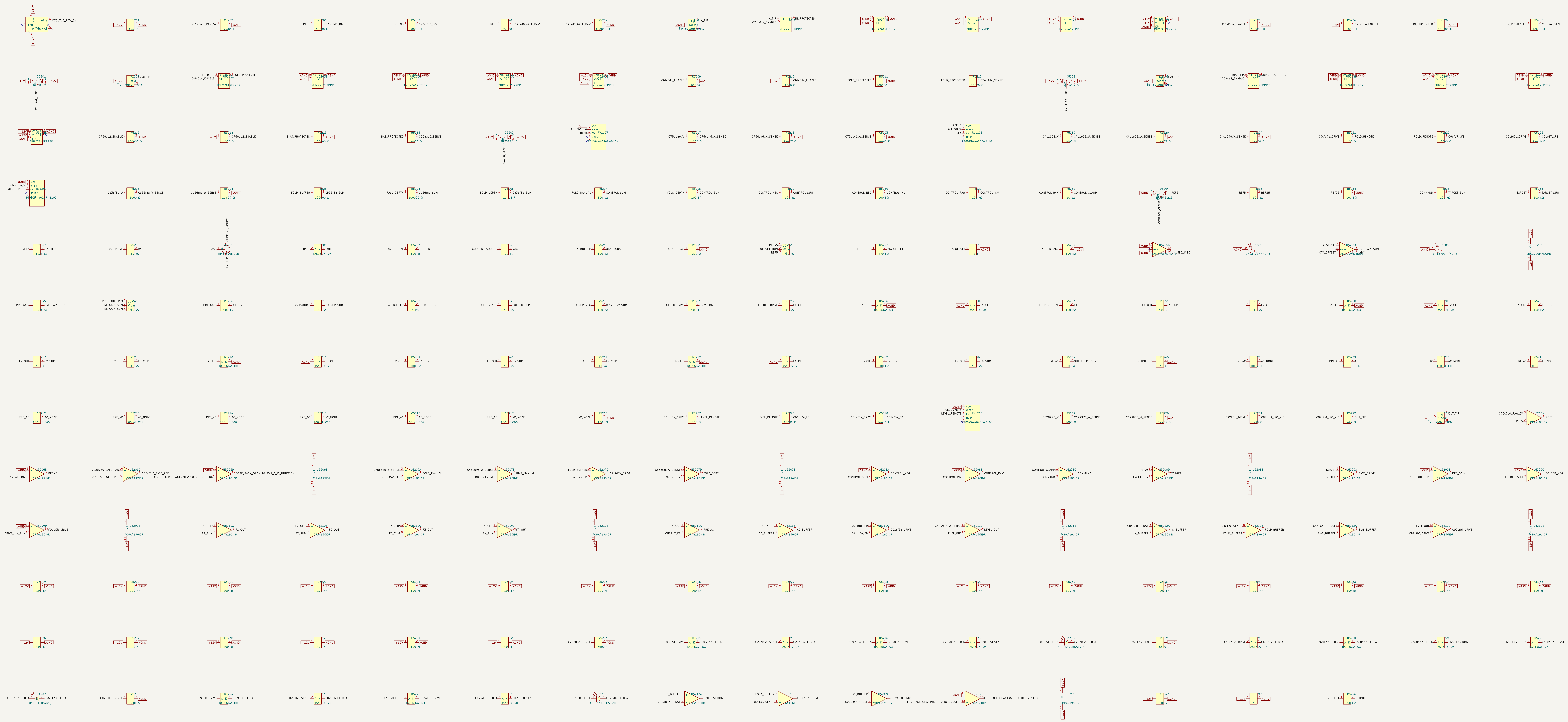
Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 28/36



Sheet: /W2/
File: wavefolder.kicad_sch

Title:

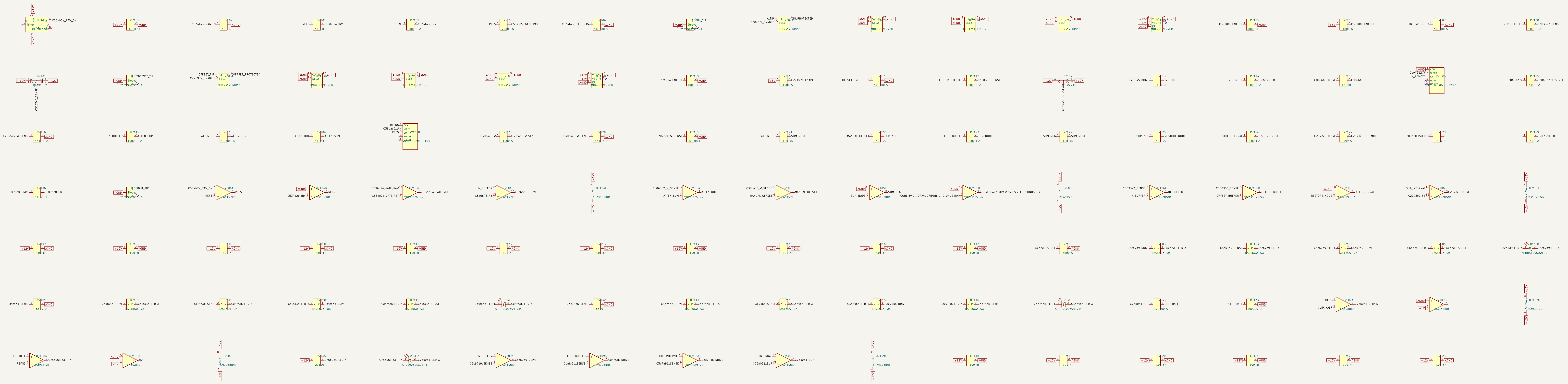
Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 29/36



Sheet: /W1/
File: wavefolder.kicad_sch

Title:

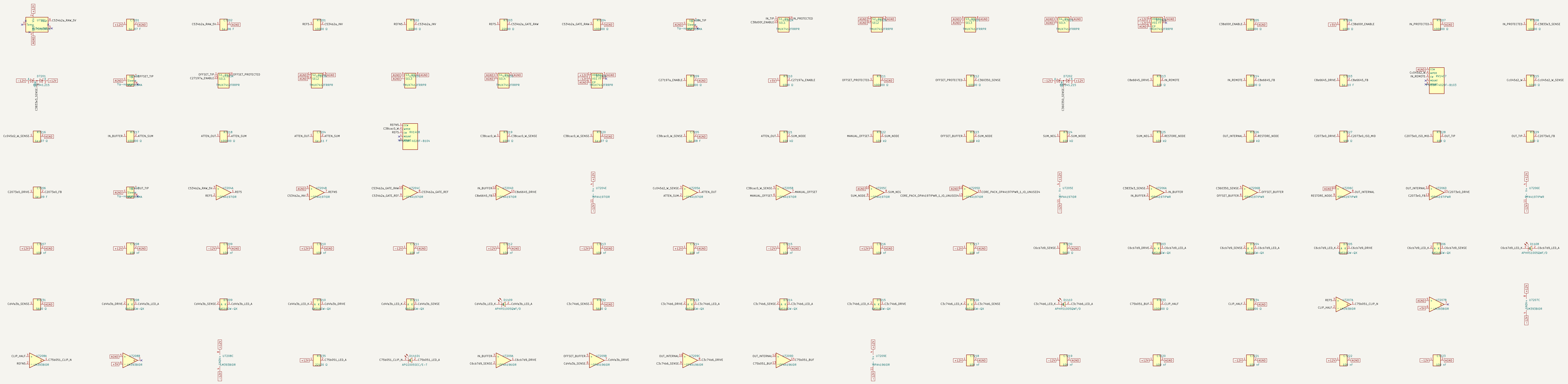
Size: A0 Date:
KiCad E.D.A. 10.0.6
Rev:
Id: 30/36



Sheet: /A01/
File: offset.kicad_sch

Title:

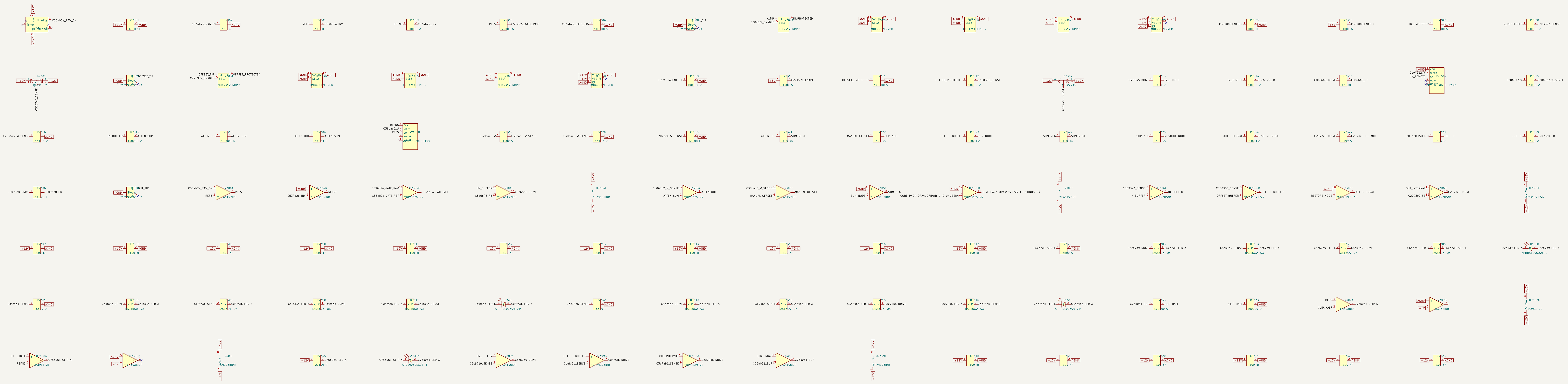
Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 31/36


U7201 REF5050AIDR +12V AGND Vin Vout Temp GND Trim/NR C534b2a_RAW_5V
C7201 +12V AGND 1e-07 F
C7202 C534b2a_RAW_5V AGND 1e-06 F
R7201 REF5 C534b2a_INV 10000 Ω
R7202 REFN5 C534b2a_INV 10000 Ω
R7203 REF5 C534b2a_GATE_RAW 22000 Ω
R7204 C534b2a_GATE_RAW AGND 100000 Ω
AGND Tip Sleeve Tip-normal TN IN_TIP WQP518MA
U7202A S1 D1 SEL1 IN_TIP C38d00f_ENABLE IN_PROTECTED TMUX7412FRRPR
U7202B S2 D2 SEL2 AGND TMUX7412FRRPR
U7202C S3 D3 SEL3 AGND TMUX7412FRRPR
U7202D S4 D4 SEL4 AGND TMUX7412FRRPR
U7202E VDD GND VSS FF EP +12V −12V AGND TMUX7412FRRPR
R7205 C38d00f_ENABLE AGND 100000 Ω
R7206 +5V C38d00f_ENABLE 1000 Ω
R7207 IN_PROTECTED AGND 100000 Ω
R7208 IN_PROTECTED C3833e3_SENSE 10000 Ω
D7201 BAT54S,215 −12V +12V A K COM C3833e3_SENSE
AGND Tip Sleeve Tip-normal TN OFFSET_TIP WQP518MA
U7203A S1 D1 SEL1 OFFSET_TIP C27197a_ENABLE OFFSET_PROTECTED TMUX7412FRRPR
U7203B S2 D2 SEL2 AGND TMUX7412FRRPR
U7203C S3 D3 SEL3 AGND TMUX7412FRRPR
U7203D S4 D4 SEL4 AGND TMUX7412FRRPR
U7203E VDD GND VSS FF EP +12V −12V AGND TMUX7412FRRPR
R7209 C27197a_ENABLE AGND 100000 Ω
R7210 +5V C27197a_ENABLE 1000 Ω
R7211 OFFSET_PROTECTED AGND 100000 Ω
R7212 OFFSET_PROTECTED C360350_SENSE 10000 Ω
D7202 BAT54S,215 −12V +12V A K COM C360350_SENSE
R7213 C8e6645_DRIVE IN_REMOTE 100 Ω
R7214 IN_REMOTE C8e6645_FB 10000 Ω
C7203 C8e6645_DRIVE C8e6645_FB 1e-10 F
RV1407 CCW WIPER CW MOUNT AGND Cc045d2_W IN_REMOTE PTV09A-4020F-B103
R7215 Cc045d2_W Cc045d2_W_SENSE 1000 Ω
R7216 Cc045d2_W_SENSE AGND 1e+07 Ω
R7217 IN_BUFFER ATTEN_SUM 100000 Ω
R7218 ATTEN_OUT ATTEN_SUM 100000 Ω
C7204 ATTEN_OUT ATTEN_SUM 1e-11 F
RV1408 CCW WIPER CW MOUNT REFN5 C38cac0_W REF5 PTV09A-4020F-B104
R7219 C38cac0_W C38cac0_W_SENSE 1000 Ω
R7220 C38cac0_W_SENSE AGND 1e+07 Ω
C7205 C38cac0_W_SENSE AGND 1e-08 F
R7221 ATTEN_OUT SUM_NODE 100 kΩ
R7222 MANUAL_OFFSET SUM_NODE 100 kΩ
R7223 OFFSET_BUFFER SUM_NODE 100 kΩ
R7224 SUM_NEG SUM_NODE 100 kΩ
R7225 SUM_NEG RESTORE_NODE 100 kΩ
R7226 OUT_INTERNAL RESTORE_NODE 100 kΩ
R7227 C2073e0_DRIVE C2073e0_ISO_MID 499 Ω
R7228 C2073e0_ISO_MID OUT_TIP 499 Ω
R7229 OUT_TIP C2073e0_FB 100 Ω
C7206 C2073e0_DRIVE C2073e0_FB 1e-09 F
AGND Tip Sleeve Tip-normal TN OUT_TIP WQP518MA
U7204A C534b2a_RAW_5V REF5 REF5 OPA4197IDR
U7204B AGND C534b2a_INV REFN5 OPA4197IDR
U7204C C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF OPA4197IDR
U7204D IN_BUFFER C8e6645_FB C8e6645_DRIVE OPA4197IDR
U7204E +12V −12V V+ V− OPA4197IDR
U7205A Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT OPA4197IDR
U7205B C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET OPA4197IDR
U7205C AGND SUM_NODE SUM_NEG OPA4197IDR
U7205D AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 OPA4197IDR
U7205E +12V −12V V+ V− OPA4197IDR
U7206A C3833e3_SENSE IN_BUFFER IN_BUFFER OPA4197IPWR
U7206B C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER OPA4197IPWR
U7206C AGND RESTORE_NODE OUT_INTERNAL OPA4197IPWR
U7206D OUT_INTERNAL C2073e0_FB C2073e0_DRIVE OPA4197IPWR
U7206E +12V −12V V+ V− OPA4197IPWR
C7207 +12V AGND 100 nF
C7208 +12V AGND 100 nF
C7209 −12V AGND 100 nF
C7210 +12V AGND 100 nF
C7211 −12V AGND 100 nF
C7212 +12V AGND 100 nF
C7213 −12V AGND 100 nF
C7214 +12V AGND 100 nF
C7215 −12V AGND 100 nF
C7216 +12V AGND 100 nF
C7217 −12V AGND 100 nF
R7230 C6cb7d9_SENSE AGND 5600 Ω
D7203 C6cb7d9_DRIVE C6cb7d9_LED_A BAS16GW-QX
D7204 C6cb7d9_SENSE C6cb7d9_LED_A BAS16GW-QX
D7205 C6cb7d9_LED_K C6cb7d9_DRIVE BAS16GW-QX
D7206 C6cb7d9_LED_K C6cb7d9_SENSE BAS16GW-QX
D1408 C6cb7d9_LED_K C6cb7d9_LED_A APHHS1005QWF/D
R7231 Cd4fa3b_SENSE AGND 5600 Ω
D7208 Cd4fa3b_DRIVE Cd4fa3b_LED_A BAS16GW-QX
D7209 Cd4fa3b_SENSE Cd4fa3b_LED_A BAS16GW-QX
D7210 Cd4fa3b_LED_K Cd4fa3b_DRIVE BAS16GW-QX
D7211 Cd4fa3b_LED_K Cd4fa3b_SENSE BAS16GW-QX
D1409 Cd4fa3b_LED_K Cd4fa3b_LED_A APHHS1005QWF/D
R7232 C3c74b6_SENSE AGND 5600 Ω
D7213 C3c74b6_DRIVE C3c74b6_LED_A BAS16GW-QX
D7214 C3c74b6_SENSE C3c74b6_LED_A BAS16GW-QX
D7215 C3c74b6_LED_K C3c74b6_DRIVE BAS16GW-QX
D7216 C3c74b6_LED_K C3c74b6_SENSE BAS16GW-QX
D1410 C3c74b6_LED_K C3c74b6_LED_A APHHS1005QWF/D
R7233 C75b051_BUF CLIP_HALF 100000 Ω
R7234 CLIP_HALF AGND 100000 Ω
U7207A REF5 CLIP_HALF C75b051_CLIP_N LM393BIDR
U7207B AGND +5V LM393BIDR
U7207C +12V −12V V+ GND LM393BIDR
U7208A CLIP_HALF REFN5 C75b051_CLIP_N LM393BIDR
U7208B AGND +5V LM393BIDR
U7208C +12V −12V V+ GND LM393BIDR
R7235 +12V C75b051_LED_A 22100 Ω
D14101 C75b051_CLIP_N C75b051_LED_A APG1005SEC/E-T
U7209A IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE OPA4196IDR
U7209B OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE OPA4196IDR
U7209C OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE OPA4196IDR
U7209D OUT_INTERNAL C75b051_BUF C75b051_BUF OPA4196IDR
U7209E +12V −12V V+ V− OPA4196IDR
C7218 +12V AGND 100 nF
C7219 −12V AGND 100 nF
C7220 +12V AGND 100 nF
C7221 −12V AGND 100 nF
C7222 +12V AGND 100 nF
C7223 −12V AGND 100 nF

Sheet: /A02/
File: offset.kicad_sch

Title:

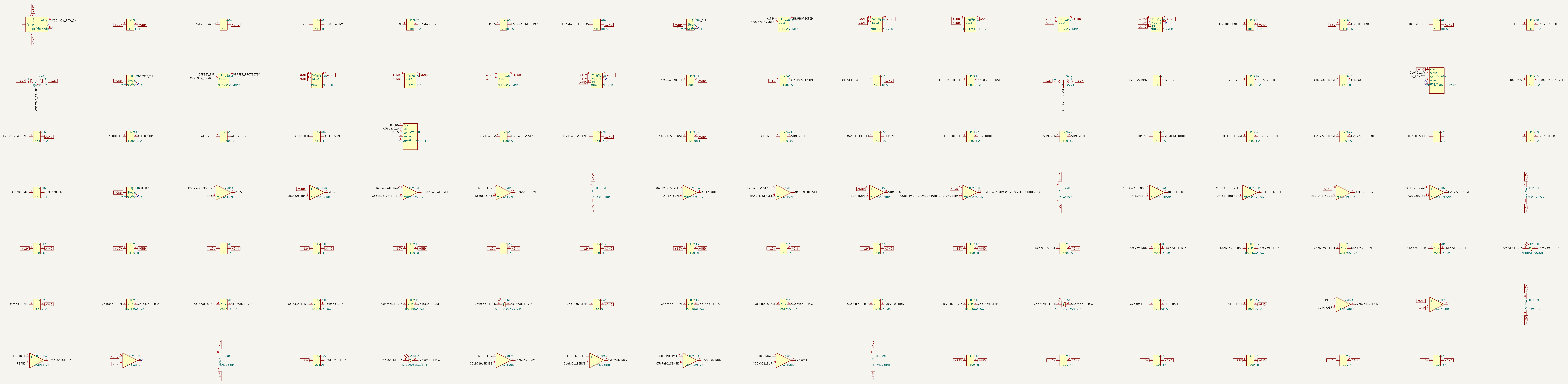
Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 32/36


U7301 REF5050AIDR Vin Vout Temp Trim/NR GND +12V AGND C534b2a_RAW_5V
C7301 1e-07 F +12V AGND
C7302 1e-06 F C534b2a_RAW_5V AGND
R7301 10000 Ω REF5 C534b2a_INV
R7302 10000 Ω REFN5 C534b2a_INV
R7303 22000 Ω REF5 C534b2a_GATE_RAW
R7304 100000 Ω C534b2a_GATE_RAW AGND
J1508 Tip Sleeve TN Tip-normal WQP518MA AGND IN_TIP
U7302A TMUX7412FRRPR S1 D1 SEL1 IN_TIP C38d00f_ENABLE IN_PROTECTED
U7302B TMUX7412FRRPR S2 D2 SEL2 AGND
U7302C TMUX7412FRRPR S3 D3 SEL3 AGND
U7302D TMUX7412FRRPR S4 D4 SEL4 AGND
U7302E TMUX7412FRRPR VDD VSS GND FF EP +12V −12V AGND
R7305 100000 Ω C38d00f_ENABLE AGND
R7306 1000 Ω +5V C38d00f_ENABLE
R7307 100000 Ω IN_PROTECTED AGND
R7308 10000 Ω IN_PROTECTED C3833e3_SENSE
D7301 BAT54S,215 A K COM −12V +12V C3833e3_SENSE
J1509 Tip Sleeve TN Tip-normal WQP518MA AGND OFFSET_TIP
U7303A TMUX7412FRRPR S1 D1 SEL1 OFFSET_TIP C27197a_ENABLE OFFSET_PROTECTED
U7303B TMUX7412FRRPR S2 D2 SEL2 AGND
U7303C TMUX7412FRRPR S3 D3 SEL3 AGND
U7303D TMUX7412FRRPR S4 D4 SEL4 AGND
U7303E TMUX7412FRRPR VDD VSS GND FF EP +12V −12V AGND
R7309 100000 Ω C27197a_ENABLE AGND
R7310 1000 Ω +5V C27197a_ENABLE
R7311 100000 Ω OFFSET_PROTECTED AGND
R7312 10000 Ω OFFSET_PROTECTED C360350_SENSE
D7302 BAT54S,215 A K COM −12V +12V C360350_SENSE
R7313 100 Ω C8e6645_DRIVE IN_REMOTE
R7314 10000 Ω IN_REMOTE C8e6645_FB
C7303 1e-10 F C8e6645_DRIVE C8e6645_FB
RV1507 PTV09A-4020F-B103 CCW WIPER CW MOUNT AGND Cc045d2_W IN_REMOTE
R7315 1000 Ω Cc045d2_W Cc045d2_W_SENSE
R7316 1e+07 Ω Cc045d2_W_SENSE AGND
R7317 100000 Ω IN_BUFFER ATTEN_SUM
R7318 100000 Ω ATTEN_OUT ATTEN_SUM
C7304 1e-11 F ATTEN_OUT ATTEN_SUM
RV1508 PTV09A-4020F-B104 CCW WIPER CW MOUNT REFN5 C38cac0_W REF5
R7319 1000 Ω C38cac0_W C38cac0_W_SENSE
R7320 1e+07 Ω C38cac0_W_SENSE AGND
C7305 1e-08 F C38cac0_W_SENSE AGND
R7321 100 kΩ ATTEN_OUT SUM_NODE
R7322 100 kΩ MANUAL_OFFSET SUM_NODE
R7323 100 kΩ OFFSET_BUFFER SUM_NODE
R7324 100 kΩ SUM_NEG SUM_NODE
R7325 100 kΩ SUM_NEG RESTORE_NODE
R7326 100 kΩ OUT_INTERNAL RESTORE_NODE
R7327 499 Ω C2073e0_DRIVE C2073e0_ISO_MID
R7328 499 Ω C2073e0_ISO_MID OUT_TIP
R7329 100 Ω OUT_TIP C2073e0_FB
C7306 1e-09 F C2073e0_DRIVE C2073e0_FB
J1510 Tip Sleeve TN Tip-normal WQP518MA AGND OUT_TIP
U7304A OPA4197IDR C534b2a_RAW_5V REF5
U7304B OPA4197IDR AGND C534b2a_INV REFN5
U7304C OPA4197IDR C534b2a_GATE_RAW C534b2a_GATE_REF
U7304D OPA4197IDR IN_BUFFER C8e6645_FB C8e6645_DRIVE
U7304E OPA4197IDR V+ V− +12V −12V
U7305A OPA4197IDR Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT
U7305B OPA4197IDR C38cac0_W_SENSE MANUAL_OFFSET
U7305C OPA4197IDR AGND SUM_NODE SUM_NEG
U7305D OPA4197IDR AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4
U7305E OPA4197IDR V+ V− +12V −12V
U7306A OPA4197IPWR C3833e3_SENSE IN_BUFFER
U7306B OPA4197IPWR C360350_SENSE OFFSET_BUFFER
U7306C OPA4197IPWR AGND RESTORE_NODE OUT_INTERNAL
U7306D OPA4197IPWR OUT_INTERNAL C2073e0_FB C2073e0_DRIVE
U7306E OPA4197IPWR V+ V− +12V −12V
C7307 100 nF +12V AGND
C7308 100 nF +12V AGND
C7309 100 nF −12V AGND
C7310 100 nF +12V AGND
C7311 100 nF −12V AGND
C7312 100 nF +12V AGND
C7313 100 nF −12V AGND
C7314 100 nF +12V AGND
C7315 100 nF −12V AGND
C7316 100 nF +12V AGND
C7317 100 nF −12V AGND
R7330 5600 Ω C6cb7d9_SENSE AGND
D7303 BAS16GW-QX A K C6cb7d9_DRIVE C6cb7d9_LED_A
D7304 BAS16GW-QX A K C6cb7d9_SENSE C6cb7d9_LED_A
D7305 BAS16GW-QX A K C6cb7d9_LED_K C6cb7d9_DRIVE
D7306 BAS16GW-QX A K C6cb7d9_LED_K C6cb7d9_SENSE
D1508 APHHS1005QWF/D C6cb7d9_LED_K C6cb7d9_LED_A
R7331 5600 Ω Cd4fa3b_SENSE AGND
D7308 BAS16GW-QX A K Cd4fa3b_DRIVE Cd4fa3b_LED_A
D7309 BAS16GW-QX A K Cd4fa3b_SENSE Cd4fa3b_LED_A
D7310 BAS16GW-QX A K Cd4fa3b_LED_K Cd4fa3b_DRIVE
D7311 BAS16GW-QX A K Cd4fa3b_LED_K Cd4fa3b_SENSE
D1509 APHHS1005QWF/D Cd4fa3b_LED_K Cd4fa3b_LED_A
R7332 5600 Ω C3c74b6_SENSE AGND
D7313 BAS16GW-QX A K C3c74b6_DRIVE C3c74b6_LED_A
D7314 BAS16GW-QX A K C3c74b6_SENSE C3c74b6_LED_A
D7315 BAS16GW-QX A K C3c74b6_LED_K C3c74b6_DRIVE
D7316 BAS16GW-QX A K C3c74b6_LED_K C3c74b6_SENSE
D1510 APHHS1005QWF/D C3c74b6_LED_K C3c74b6_LED_A
R7333 100000 Ω C75b051_BUF CLIP_HALF
R7334 100000 Ω CLIP_HALF AGND
U7307A LM393BIDR REF5 CLIP_HALF C75b051_CLIP_N
U7307B LM393BIDR AGND +5V
U7307C LM393BIDR GND V+ +12V −12V
U7308A LM393BIDR CLIP_HALF REFN5 C75b051_CLIP_N
U7308B LM393BIDR AGND +5V
U7308C LM393BIDR GND V+ +12V −12V
R7335 22100 Ω +12V C75b051_LED_A
D15101 APG1005SEC/E-T C75b051_CLIP_N C75b051_LED_A
U7309A OPA4196IDR IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE
U7309B OPA4196IDR OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE
U7309C OPA4196IDR OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE
U7309D OPA4196IDR OUT_INTERNAL C75b051_BUF
U7309E OPA4196IDR V+ V− +12V −12V
C7318 100 nF +12V AGND
C7319 100 nF −12V AGND
C7320 100 nF +12V AGND
C7321 100 nF −12V AGND
C7322 100 nF +12V AGND
C7323 100 nF −12V AGND

Sheet: /A03/
File: offset.kicad_sch

Title:

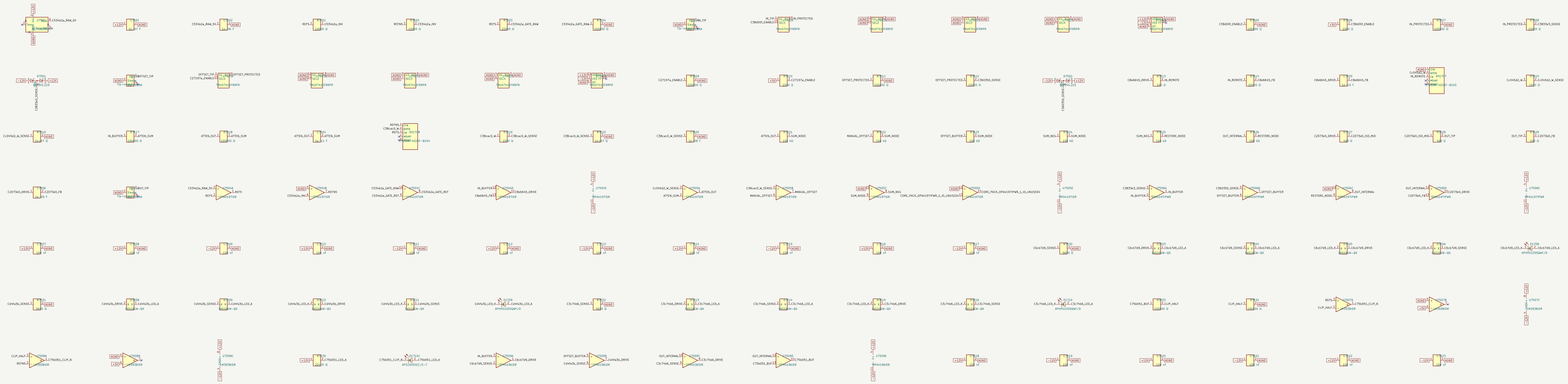
Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 33/36



Sheet: /A04/
File: offset.kicad_sch

Title:

Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 34/36

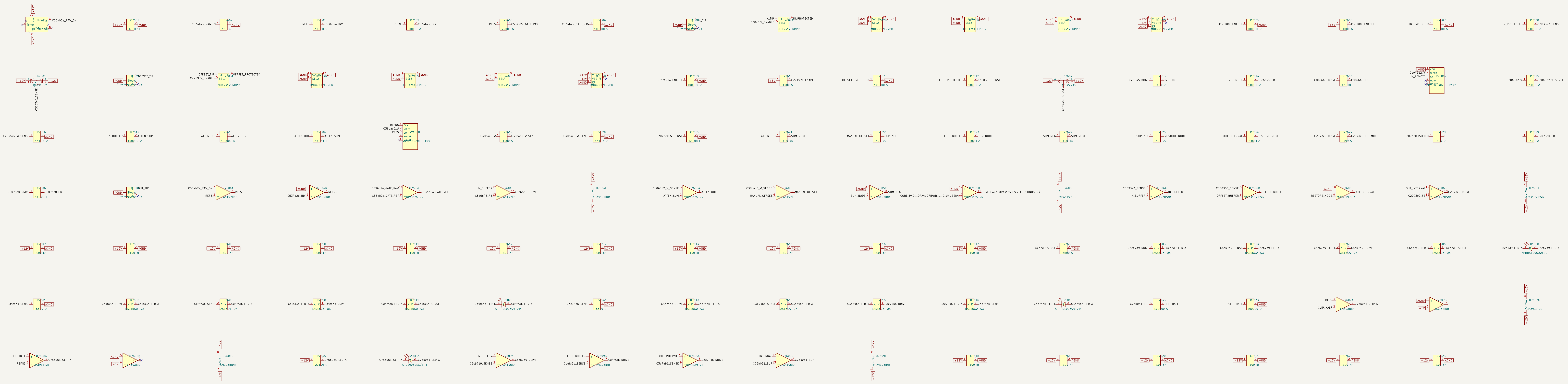


Sheet: /A05/
File: offset.kicad_sch

Title:

Size: A0 Date:
KiCad E.D.A. 10.0.6

Rev:
Id: 35/36



Sheet: /A06/
File: offset.kicad_sch

Title:

Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 36/36